



The cross-border effects of bank capital regulation[☆]

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ABSTRACT

We study the international coordination of bank capital requirements under a host-country rule: the requirement depends on where the borrower, not the bank, is located. In such a regime, countries compete for scarce bank equity capital. Raising a country's requirement may generate bank capital outflows as well as inflows. We pin down the condition for the sign of the capital flow and the associated externality, and highlight the policy implications. Absent collaboration, overshooting is likely: individual countries have an incentive to increase Basel III's Counter-Cyclical Capital Buffer too much in good times and cut it too much in bad times.

1. Introduction

International standards for bank capital regulation have evolved over the past four decades from a simple, common 8% minimum capital requirement to a broad, complex toolkit. Macroprudential considerations have been a driving force behind this evolution. Prudential risks and macroeconomic cycles differ across countries, which suggests the need for heterogeneous and time-varying capital requirements. Under the current regime (Basel III) the stance of macroprudential regulation is effectively set at the discretion of national regulators via so-called countercyclical buffers (CCyBs). These buffers apply to all bank loans made in the regulator's jurisdiction, irrespective of which jurisdiction

the bank belongs to. Accordingly, we study a model in which capital requirements follow a host-country rule: they are set by the country in which lending takes place.

The existing literature has studied home-country rules under which the requirement is set by a bank's home regulator regardless of where lending takes place. In these circumstances, international competition for credit market share within a country is a key driver of strategic incentives: Regulators can give an advantage to banks in their jurisdiction by cutting capital requirements as this allows them to operate at a cheaper cost than banks from other jurisdictions. So, to the extent that regulators care about the profits of the banks they regulate, they

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¹ Typical concerns associated with higher capital requirements in a given country go from impairing the competitiveness of the domestic financial system (Osborne, 2015) to a reduction in domestic banks' foreign exposures, therefore impairing the functioning of foreign financial systems (de Guindos, 2019), and to cross-border relocation of risk-shifting activities (ESRB, 2018), page 90).

have an incentive to undercut one another in order for their banks to steal market share (Dell'Ariccia and Marquez, 2006). Common minima eliminate this market-share externality. But, as we have outlined, national regulators have discretion over the stance of macroprudential requirements, and regulators and policymakers regularly express concerns about the international spillovers of capital requirements in the current regime. Yet there is little agreement on what the relevant externalities are and a formal framework for assessing them is needed.¹

We find that, under a host-country rule, what matters is not competition for market share but competition for bank capital. Key is how much equity capital is allocated to lending in different jurisdictions. Changes in capital requirements alter this allocation and effectively generate bank equity capital flows. We propose a model to study such capital flows and their implications for strategic interactions between regulators. The model has two dates and two countries (*Home* and *Foreign*), in which banks finance loans with a mix of insured deposits and equity capital. Risk-shifting incentives make banks endogenously specialise in lending in either country (so, we refer to them as Home or Foreign banks). These banks face capital requirements set by the regulator of the country in which the lending takes place. Bank equity capital is mobile and there is global competition for it.

Our contribution is threefold: First, we show that, perhaps against conventional wisdom, an increase in the capital requirement in a country does not necessarily generate outflows of bank equity capital – *inflows* are possible too. We pin down the conditions under which either case occurs. Second, as bank equity capital is scarce, changes to capital requirements in a country impose, through capital flows, an externality onto the other country. We show that this *capital flow externality* is central to the incentive for national regulators to deviate from a collaborative optimum. If higher requirements generate positive externality, there is an incentive to deviate downwards, i.e., to undercut the other country (and vice versa if the externality is negative). Third, we point out the implications for the coordination of macroprudential capital regulation. In particular, under a host-country rule and absent coordination, macroprudential capital requirements are likely to be raised too much in good times and cut too much in troubled times, when bank equity capital is particularly scarce.

To understand these results, let us expose the main mechanism of the model, starting from the perspective of a single country, Home. There are diminishing returns to capital in the economy, which translates into a downward sloping demand for loans. Imagine Home bank equity capital is fixed. Then, an increase in capital requirement will contract lending and raise its marginal return. If initial quantities are relatively high, this fall in lending increases profitability. The intuition is the same as uncompetitive firms in a product market boosting their profits by reducing quantities. However, if initial quantities are relatively low a fall in lending reduces profits; in the same way that collusive firms would lose profit if they reduced quantities past the point a monopolist would choose. Due to this mechanism, the Home banks' profits and their return on equity are hump-shaped in the level and lending, and hence, in the capital requirement. If the capital requirement is initially below the return maximising level, then an increase will raise equity returns (and, vice versa, it will lower returns if we start above the return maximising level).

Now take a cross-border perspective. In our model, banks issue equity competitively to investors who supply it at a cost increasing in the total amount raised across countries. Hence, there is a global supply curve for equity. Investors face no constraints on where they invest and, in equilibrium, the return on bank equity is equal across countries (and to the cost of raising equity). Imagine a change in capital requirements at Home that increases the Home bank return on equity. Home banks respond by demanding more equity. This shifts the equilibrium along the global supply curve. Hence, Foreign banks face an inward shift in their residual supply curve. There is a higher equilibrium cost of equity,

which results in a contraction in lending in Foreign.² Such a spillover is at the heart of our analysis.

We interpret this spillover as a capital flow. To understand why, consider an extreme case where there is a fixed global supply of capital. All capital flowing into Home must flow out from Foreign; we have a 100% spillover. Conversely, if the global supply is perfectly elastic, all capital flowing to Home will be newly raised capital and there is no spillover. For intermediate cases, part of the equity capital increase in Home can be seen as being siphoned off from Foreign.

Intuitively, the direction of the equity capital flow following an increase in the Home capital requirement depends on whether the Home bank return on equity increases or decreases. If there is an increase, equity capital flows into Home, if there is a decrease, capital flows out. We can now link back to the single-country perspective described above: here too, whether or not the return on equity increases depends precisely on whether the initial capital requirement is above or below the return maximising level. If Home capital requirements are below the return maximising level this implies that the initial level of lending is relatively high. Then a tighter requirement restrains the quantity of lending at Home which raises profits and hence the Home bank return on equity. So capital flows out of Foreign banks. (Vice versa, if initial requirement is high, lending is low, tighter requirements lower Home returns and cause a capital flow to Foreign banks).

Having characterised this mechanism for a given set of capital requirements, we then turn to a policy game in which we endogenise the requirements. Regulators seek to maximise net output subject to deadweight losses from financial instability. Bank equity capital is socially valuable as it alleviates such losses. But banks prefer to operate with maximum leverage so there is role for capital regulation. As we have seen, regulation has crossborder effects which generates strategic incentives. We compare the Nash equilibrium to collaboration. Nash regulators effectively compete for bank equity capital.

When a tighter capital requirement attracts equity capital to Home, some of it is siphoned off from Foreign. There is a negative externality; part of the cost of tighter regulation is borne abroad. In contrast, if tighter requirements lead to an outflow from Home, Foreign benefits. Under Nash competition, regulators ignore these effects. As a result, whether they have an incentive to deviate upwards or downwards from the collaborative outcome depends on the sign of the capital flows the deviation would generate.

Which case arises depends on parameter values. When equity capital is abundant, collaboration can achieve a high level of lending while preserving financial stability with relatively tight requirements. Given that the collaborative level of lending is high, we are in the region where marginally increasing the capital requirement raises profitability. Tighter regulation therefore generates inflows, the incentive to deviate is upward, and the Nash equilibrium corresponds to a race-to-the-top scenario. In contrast, when equity capital is relatively scarce, the level of lending is low. Looser requirements therefore generate inflows and the incentive is to undercut the other regulator. The Nash equilibrium of the policy game corresponds to a race-to-the-bottom scenario.

Our analysis suggests that competing regulators will raise the CCyB too high in normal times (when bank capital is not too scarce).³ In

² This can be interpreted as the representative Home bank growing its equity capital while the Foreign bank shrinks. An alternative way to interpret this is through entry. Banks exit the Foreign credit market and enter the Home market. Since there is free entry, perfect competition and banks have no fixed costs, either interpretation is valid.

³ As we discuss in Section 5, the empirical literature is generally consistent with the notion that banks respond, most of the time, to increases in capital requirements by partially raising more equity (as well as adjusting assets). In our model, this happens if and only if higher capital requirements generate capital inflows. This supports our interpretation of normal times being those where capital is not too scarce and where the level of lending is larger than the level that maximises investor revenue.

this case, there are gains from coordinating on more modest raises. However, if bank capital is very scarce (think of bad or troubled times, for instance after a big negative shock to bank equity capital), then competing regulators will have an incentive to cut the CCyB by more than what collaboration requires.

Ceteris paribus, a larger weight put on financial stability in the welfare function implies a lower collaborative level of lending and therefore a higher likelihood that it is in the region where the incentive to deviate is downward. Hence, the race-to-the-bottom scenario becomes more likely after a negative shock that would raise the potential cost of financial instability (for instance, a shock to fiscal capacity).

The reference paper for a home-country rule is Dell'Ariccia and Marquez (2006). To illustrate how, in such a regime, regulators' strategic interactions fundamentally differ from those in the one we study, it is best to highlight a key mechanism in their paper. They have a representative bank based in each of two countries, but both banks operate in both countries. Each bank has a fixed amount of equity capital and faces the capital requirement imposed by their country of origin. A key point is that a decrease in capital requirement by the Home regulator decreases the cost of capital for the Home-based bank, which gives it a competitive advantage in both markets and allows it to grab market share from its Foreign competitor. This is an externality that naturally gives incentives for countries to undercut one another. Adopting a reciprocity regime kills such a market-share externality.⁴

In a similar international context, a series of papers study the interaction between capital regulation and other policy levers. Acharya (2003) looks at how discretion in resolution regimes can undermine the benefit of coordination in capital regulation. Morrison and White (2009) examine the link between banking regulation and supervisory quality. In their set up, capital requirements are a substitute to the regulator's ability to distinguish sound banks from weak ones. Competition among regulators creates a selection effect: high quality banks prefer to be chartered by high ability regulators, which also set lower capital requirements than low ability regulators.⁵

In addition, a sequence of papers have focused on international coordination of bank supervision, rather than capital requirements. Carletti et al. (2020) and Colliard both consider the role of central and local supervision when local supervisors have informational advantages but neglect cross-border externalities. Similarly, Calzolari and Loranth (2011) and Calzolari et al. (2018) consider how the presence of multinational banks alter supervisory incentives.

At a more general level, Korinek (2016) identifies conditions under which there is scope for international policy collaboration. The relevant one in our paper is the violation of the Tinbergen Principle: Regulators have a single tool (the capital requirement) and face a policy trade off (e.g. between economic activity and financial stability). If regulators could, for instance, directly and costlessly subsidise the allocation of bank equity to domestic lending, then the trade off could disappear, and so could the gains from collaboration.

In practice, fiscal policy is mostly separated from macro-prudential policy. How these policies should be coordinated within and between countries is a complex, multifaceted question. Government guarantees generate implicit subsidies that can distort bank behaviour and have direct fiscal implications that can spillover across borders when banks fail. Faia and Weder (2016), Bolton and Oehmke (2018) and Segura and Vicente (2019) study how the resolution of banks should be coordinated between countries. Moreover, monetary policy can directly

affect bank profitability, which may also call for coordination across borders.⁶ Modelling all these different aspects of policy is beyond the scope of this paper, we focus on a prudential policymaker that only controls the capital requirement.

Finally, even though our main focus is on strategic interactions between national regulators, other dimensions of our analysis directly relate to previous literature on capital requirements. Commonalities include moral hazard due to government guarantees (Kareken and Wallace, 1978), incentives for banks to specialise (e.g., Martinez-Miera and Suarez (2014), Harris et al. (2023), Bahaj and Malherbe (2020) and Malherbe and McMahon (2024)), and the study of optimal capital regulation in the face of policy trade offs (e.g., Begenau (2020), Malherbe (2020) and Elenev et al. (2021)).

2. The model

The model has two dates: 0 and 1. Decisions are made at date 0. At date 1, all stochastic variables are realised, and production and consumption take place.

We consider two sovereign countries: Home and Foreign. There is a single, tradeable good which can be consumed or used as physical capital in production, in which case the goods depreciate fully. This good is also the numeraire.

In each country, there is a representative firm, a representative household, a representative investor, and a regulator. There is also a mass of banks that can operate in both countries. We describe here the details of the environment in the Home country. The Foreign country has the same environment (although we do not necessarily impose symmetry in parameter values); foreign variables are marked with '.

Agents and preferences. Private agents only value date-1 consumption, are risk neutral, and act competitively. We define and discuss regulator preferences starting in Section 4.

Firms and technology. The representative firm operates a Cobb–Douglas production technology: $Ak^\alpha l^{1-\alpha}$, where k is physical capital, l is labour, $0 < \alpha < 1$, and $A \geq 0$ is a random variable that captures TFP. The firm invests in capital at date 0. At date 1, A realises and the firm hires labour and produces.

We normalise $\mathbb{E}[A] = 1$ and assume that A is distributed over $[0, A^H]$, with a corresponding density function $g(A)$, which is smooth, unless otherwise specified. Our analysis does not require to impose any specific structure on the dependence between A and A' but we assume that the joint distribution has full support over $[0, A^H] \times [0, A'^H]$.

The firm starts penniless and borrows from banks to invest in physical capital. In equilibrium, it makes zero profits in all states. We therefore abstract from firm ownership.

All agents have access to a riskless storage technology with a zero rate of return.

Households. The representative household is endowed with one unit of labour, which it supplies inelastically (and, for simplicity, without disutility) at date 1. It also has an endowment of goods at date 0, which it initially allocates between insured bank deposits and the storage technology. We assume that this endowment is sufficiently large that the storage technology is always used in equilibrium. This pins down a households' opportunity cost of funds of unity.

⁴ The Dell'Ariccia and Marquez (2006) model is more involved and also embeds a financial stability externality, which reinforces the market share externality. We also allow for additional externalities but only discuss them in Section 5 since they are not the focus of our analysis.

⁵ Other examples include, Buck and Schliephake (2013) and Gersbach et al. (2020), which respectively focus on capital regulation interactions with supervision intensity and fiscal policy.

⁶ From that angle, our paper relates to the wider literature on monetary policy coordination and currency wars (e.g., Hamada (1976), Corsetti and Pesenti (2001), Caballero et al. (2021) and Blanchard (2021)).

Banks. Banks start penniless, issue equity (protected by limited liability), take deposits that are insured at no cost, and lend to firms subject to a capital requirement constraint that we define below. Banks can potentially lend in both countries. There is free entry, and equity and deposits can be raised globally (i.e. from investors and households in either country). Banks can freely choose their country of incorporation, which determines which taxpayer insures its deposits. Deposit insurance is funded by ex-post lump-sum taxes on households.

Capital requirements: host country rule. Loans made in Home face a capital requirement $\gamma \in (0, 1)$ that is set by the Home regulator. Likewise, the Foreign regulator sets $\gamma' \in (0, 1)$, which applies to loans made in Foreign. There is only one type of regulatory capital for banks: equity. To avoid confusion with physical capital, we henceforth refer to it as bank *equity capital*.

Consider a bank i and denote x_i and x'_i the quantity it lends in Home and Foreign, respectively, and denote its equity capital n_i . This bank faces the following constraint,

$$n_i \geq \gamma x_i + \gamma' x'_i. \quad (1)$$

In Section 3, γ and γ' are treated as exogenous parameters, but from Section 4 they are the choice variables of a policy game.

Banks could also invest in storage which would carry a zero capital requirement. However, raising deposits to invest them in storage would have no impact on the bank's net cash flows, so we abstract from this in what follows.

Investors and the supply of bank equity capital. That bank equity capital is scarce is a key ingredient of our analysis. For tractability reasons, many papers in the literature (e.g., Dell'Ariccia and Marquez (2006)) assume that the supply is fixed. However, that bank equity scarcity depends on the state of the economy seems the premise for the need of time varying capital requirements (Kashyap and Stein (2004), Kashyap et al. (2008), Malherbe (2020)). So, we adopt a more flexible approach and, following Hellmann et al. (2000), we assume a generally upward sloping supply curve. We formalise it as follows.

Only the agents that we call *investors* can invest in bank equity. They are endowed with some initial wealth and can generate additional date-0 wealth, at a disutility cost. Specifically, to generate m units of wealth, an investor has to incur a disutility cost $m(1 + z)$, with

$$z \equiv \frac{\kappa}{2} M^2, \quad (2)$$

and where $\kappa > 0$ is a parameter and $M \geq 0$ is the global additional wealth generated by investors. Investors take the unit disutility cost as given. Generating additional wealth is costly and investors only potentially do so for the purpose of investing in bank equity, accordingly, we have:

$$M \equiv \begin{cases} 0 & N + N' \leq \omega \\ N + N' - \omega & N + N' > \omega, \end{cases}$$

where $N + N'$ is the aggregate amount invested in bank equity globally, and ω is the initial global endowment of investor wealth. Investors supply equity competitively and freely across borders. Therefore, function (2) constitutes the global bank equity inverse supply curve.

The allocation of $N + N'$ between the representative investor in each country will not play a role in our analysis (and neither does the initial split of ω). To rule out uninteresting cases where equity is fully abundant, we impose: $\omega < 2\alpha^{\frac{1}{1-\alpha}}$.

Market equilibrium. In an equilibrium: given capital requirements γ and γ' , banks and firms maximise expected pure profits (i.e., revenue minus the total costs, including capital); households and investors maximise utility (i.e. date-1 consumption less any disutility of wealth generation in date 0); and markets clear.

Preliminaries. The problems of the firms, the households, and the investors are trivial. Price taking behaviour implies that, in equilibrium:

- Deposits promise a zero net rate of return.
- The wage is given by: $A(1 - \alpha)K^\alpha$, where K is aggregate capital. Given labour supply is normalised to 1, this corresponds to the aggregate wage bill.
- The loan gross interest rate is $A\alpha X^{\alpha-1}$, where X is aggregate lending.⁷
- Investors break even in expectation. So, in equilibrium, M is such that the expected return on bank equity equals its marginal cost, i.e., the unit disutility of generating additional wealth $(1 + z)$.

Market clearing requires:

- $K = X$.

The banks' problem is more involved and is the focus of the next section. However, parameter restrictions $\gamma, \gamma' < 1$ ensure that banks will never be 100% safe. This allows us to focus attention on equilibria where capital requirements have interesting effects.

3. Positive analysis: equity capital flows for given capital requirements

In this Section, we focus on the equilibrium behaviour of banks for a given pair of capital requirements. We treat γ and γ' as parameters, and study how a small change in γ affects the equilibrium allocation.

3.1. The banks' problem and the market equilibrium

Lemma 1. (i) Banks default with strictly positive probability in equilibrium; (ii) capital requirements are binding; and (iii) each individual bank perfectly specialises as either a lender in Home or Foreign.

Proof. All proofs are in Appendix C. $\square$

Bank shareholders break even in expectation. Depositors also break even, but when a bank defaults they do so at the taxpayer's expense. Since deposits are in effect subsidised by the taxpayer, they are seen by the bank as cheaper than equity. This is why the capital requirement binds.⁸

Government guarantees also induce banks, *ceteris paribus*, to minimise diversification. Such guarantees endow the bank with a valuable put option on their realised net worth (Merton, 1977). The value of this option is maximised when all of the bank's loans generate negative cash flows (net of repayments to depositors) in the same state of the world (Harris et al., 2023). Hence, in equilibrium, it is optimal for banks to specialise in lending in just one country.⁹

⁷ For simplicity, we consider an interest rate contingent on the realisation of TFP. Since firm defaults are costless, the realised repayment is exactly identical to what it would be in an equilibrium under a standard debt contract with face value $A^H \alpha X^{\alpha-1}$ per unit of debt.

⁸ In practice, capital requirements are not, strictly speaking, binding, as banks hold voluntary buffers to forestall violations in the event of small shocks. If these buffers are stationary, banks revert to target buffers after shocks and ultimately meet a change in requirement with an equivalent change in the capital ratio. Then, capital requirements are *essentially* binding, which is sufficient as a premise for our analysis. There is substantial empirical evidence that capital regulation influences bank decisions both at the bank-level (Gropp et al., 2019) and within bank portfolios (Behn et al., 2016). See Bahaj et al. (2016) for direct evidence supporting the essentially binding requirement hypothesis.

⁹ Given free movement of capital and risk-neutrality, the bank's problem can be separated in two: (i) optimise lending decision in each country; and (ii) choose where to operate. Lending in the two countries simultaneously is dominated as it would reduce the value of the put option derived from the guarantee.

In practical terms, this means banks prefer to lend via separate balance sheets in each country (i.e. subsidiaries) rather than lend via branches (or engage in direct cross-border lending), which would diversify risk in a single balance sheet. Accordingly, one interpretation of our setup is that banks set up holding companies that operate across borders through separate subsidiaries in each country. Alternatively, we can think of individual banks as stand-alone specialised lenders in each country. The former is closer to how banks operate in reality. However, for ease of exposition, and to save on notation, we will present our analysis using the interpretation of banks as stand-alone specialised lenders in each country.

Additionally, the market for bank equity capital is global, deposits promise a zero return in both countries, capital requirements follow a host-country rule, and there are no corporate taxes. Hence, banks are indifferent regarding their country of incorporation. This plays no role for their behaviour and so we only distinguish banks along the dimension of where they lend. We impose a tie-breaking rule that banks incorporate in, and issue deposits that are insured by, the country where they lend.

The key object: Equity capital allocated to lending. The key object in our analysis will be N , the aggregate quantity of bank equity capital allocated to banks specialising in lending at Home. A change in γ causes a reallocation of equity capital between Home (N) and Foreign (N'): effectively a bank equity capital flow. It turns out that the direction of this capital flow governs the strategic interactions between regulators. We will come on to this later. In this section we first look at the economic mechanisms that pin down the direction of the capital flow in the market equilibrium.

Bank return on equity. We consider a representative bank specialising in lending to Home firms. We denote this bank's equity n . Since capital requirements bind, the bank's lending is $x = \frac{n}{\gamma}$, and the proceeds from lending are given by $\frac{n}{\gamma} (A\alpha X^{\alpha-1})$ where the bank takes X as given. To lend an amount $\frac{n}{\gamma}$, the bank raises a total of $(1-\gamma)\frac{n}{\gamma}$ of deposits on which it pays zero interest. Aggregating across banks we have $X = \frac{N}{\gamma}$.

Define A^0 as the realisation of A such that the bank just has sufficient proceeds from its loans to make depositors whole. That is

$$A^0(N, \gamma) \equiv \frac{(1-\gamma)}{\alpha \left(\frac{N}{\gamma}\right)^{\alpha-1}}.$$

Given Lemma 1, the problem of the bank boils down to:

$$\max_{n, x \geq 0} x \underbrace{\int_{A^0(N, \gamma)}^{A^H} (\alpha A(X)^{\alpha-1} - (1-\gamma)) g(A) dA}_{\text{shareholder payout}} - \underbrace{n(1+z)}_{\text{cost of equity}}. \quad (3)$$

The second term reflects the cost of equity, where banks take z as given. Shareholders are protected by limited liability. So, the first term is their expected payout.

Shareholders have collectively invested equity capital n , hence, their expected return on equity is:

$$R(N, \gamma) \equiv \frac{1}{\gamma} \int_{A^0(N, \gamma)}^{A^H} \left(\alpha A \left(\frac{N}{\gamma} \right)^{\alpha-1} - (1-\gamma) \right) g(A) dA. \quad (4)$$

Market equilibrium. Let N^* denote an equilibrium level of Home capital.

Proposition 1. For all $\gamma, \gamma' \in (0, 1)$, there exists a unique pair $\{N^*, N'^*\}$ such that returns on equity in both countries are equal to the investors' required return. This pair is implicitly defined by:

$$R(N^*, \gamma) = R'(N'^*, \gamma') = (1 + z(N^*, N'^*)), \quad (5)$$

For intuition, note that, for a given γ , an increase in N implies more lending and therefore more aggregate physical capital. From diminishing returns, it directly follows that $R(N, \gamma)$ is decreasing in N

(and $R'(N', \gamma')$ in N'). This leads to the result since $z(N, N')$ is weakly increasing, and strictly from $N + N' = \omega$.¹⁰

In our interpretation that banks are stand-alone entities that specialise in lending in either country, international capital mobility means that they can raise equity capital from investors in either country. In the alternative bank holding company interpretation, one can think of the holding company raising equity (mainly or even exclusively) where it is incorporated, and then allocating it to its subsidiaries via its internal capital markets. Either way, Eq. (5) is the relevant indifference condition. That deposits are mobile internationally plays no role in our analysis. All results would, for instance, go through if deposits could only be raised in the country where a bank lends.

3.2. International spillovers

We now turn to how capital requirements alter the allocation of bank equity capital to either Home or Foreign.

Starting from equilibrium, we now consider the effect of marginal changes in the Home capital requirement γ . The no-arbitrage condition in Proposition 1 implicitly defines a function $N^*(\gamma, \gamma')$. From now on, functions denoted with a $*$ are evaluated at the market equilibrium arising from a couple $\{\gamma, \gamma'\}$ and hence have the two capital requirements as their arguments. For compactness, we sometimes drop function dependencies below. We represent partial derivatives with a subscript, and we write total derivatives with a $\dot{}$.

We first characterise how N^* adjusts following a change in γ .

Lemma 2. For all $\gamma, \gamma' \in (0, 1)$,

$$\frac{dN^*}{d\gamma} = R_\gamma(N^*, \gamma) \xi^*(\gamma, \gamma'), \quad (6)$$

$$\text{where } \xi^*(\gamma, \gamma') \equiv \left(\frac{\kappa M - R'_{N'}}{R_N R'_{N'} - \kappa M (R'_{N'} + R_N)} \right) \bigg|_{N^*(\gamma, \gamma'), N'^*(\gamma', \gamma)} > 0. \text{ Therefore}$$

$$\frac{dN^*}{d\gamma} \gtrless 0 \Leftrightarrow R_\gamma(N^*, \gamma) \gtrless 0.$$

Eq. (6) states that the response of N^* to γ depends on κ and on R_N and $R'_{N'}$, the sensitivity of returns to the equity capital invested in the respective countries. We will elaborate below on the role of these objects.

The second part of the lemma establishes that $\frac{dN^*}{d\gamma}$ and $R_\gamma(N^*, \gamma)$ have the same sign. The intuition is simple: a change in capital requirements will trigger an increase (decrease) in bank equity capital in Home if, and only if, this change increases (decreases), ceteris paribus, the Home bank return on equity.

By construction, additional equity capital can only be raised by Home banks from two different sources: (i) investors can supply additional wealth; (ii) there can be a flow of capital from Foreign banks to Home banks. In general, we get a combination of the two.

The function $R'(N', \gamma')$ is the inverse demand curve for N' and, hence, can be thought of as an inverse supply curve for N arising from capitals flows. Likewise, $z(N, N')$ is the inverse supply curve from investors generating additional wealth. These two curves have respective slopes $R'_{N'}$ and κM . With this in mind, the proposition that expresses the capital flow naturally follows:

Proposition 2. For all $\gamma, \gamma' \in (0, 1)$,

$$\frac{dN'^*}{d\gamma} = - \underbrace{\left(\frac{\kappa M^* - R'_{N'}}{\kappa M^* - R'_{N'}(N'^*, \gamma^*)} \right)}_{\equiv SP^*(\gamma, \gamma') \in [0, 1]} \frac{dN^*}{d\gamma}. \quad (7)$$

¹⁰ While the pair $\{N^*, N'^*\}$ is unique, the proportion of bank equity capital sourced from investors in Home or Foreign is indeterminate as investors are indifferent in equilibrium.

That $\frac{dN^*}{d\gamma}$ has the opposite sign to $\frac{dN^*}{d\gamma}$ indicates that any change in capital in Home is met by a change of opposite sign in Foreign. The term which we denote SP^* (for spillover), determines the relative proportion of these changes. Intuitively, SP^* is pinned down by the relative slopes of the implicit supply curves we mentioned above. Imagine that $M^* > 0$. A larger κ increases the spillover and, as $\kappa \rightarrow \infty$, $SP^* = 1$. In that case, any increase in N^* must correspond to an equal decrease in N'^* . Conversely, as $\kappa \rightarrow 0$, $SP^* \rightarrow 0$: a change in capital requirements at Home does not affect the equilibrium cost of bank equity capital: supply adjusts entirely through raising fresh equity capital, and there is no spillover. Similarly, if $M^* = 0$, there initially is an excess supply of investor wealth (i.e. $\omega > N'^* + N^*$) and we have $z^* = 0$. So investors can supply more capital to Home without affecting Foreign.

Remark. Strictly speaking, non-zero spillovers require that $M^* > 0$, and not necessarily that $z^* > 0$: What is needed to generate a spillover is not that capital is costly per se, but that the equilibrium cost of capital is affected by a change in capital requirements. In a model where $z^* > 0$ (say, due to a tax advantage of debt) but $M^* = 0$, there would be no spillover as a change in the capital requirement at Home would have no effect on the equilibrium cost of capital in Foreign.

3.3. The direction of international bank equity flows

Lemma 2 and Proposition 2 show that the direction of equity capital flows hinges on the sign of $R_\gamma(N^*, \gamma)$. We now discuss the economics of this sign and establish that for any γ' there is a threshold value for γ that pins down the direction of the capital flow.

3.3.1. An increase in γ can raise return on equity ($R_\gamma(N^*, \gamma) > 0$)

Given that R embeds a subsidy from a government guarantee, which is decreasing in the capital requirement, it is perhaps natural to assume that $R_\gamma(N^*, \gamma)$ is negative and higher capital requirements always generate capital outflows. However, there is another force via which increases in γ can raise bank return on equity. Raising γ decreases aggregate lending in Home, which can increase total revenues in the banking sector, much as if there had been a decrease in competition. Crucially, increased revenues translate to a higher return on equity. We first formalise this intuition with a simplified example and then come on to how things play out in our model.

A simplified problem. Consider our environment, but without uncertainty (set $A = 1$) and consider a fictitious monopolist bank, with predetermined capital N , that maximises pure profits. The monopolist's optimal level of lending, X^{mon} , is given by:

$$X^{\text{mon}} \equiv \arg \max_X \alpha X^\alpha - X - zN. \quad (8)$$

The objective is hump-shaped in X , which reflects monopoly rents: starting from low levels, pure profits increase up to X^{mon} , where they peak, and then decrease. Profits are hump-shaped in X , and so is the return on equity (for given N):

$$R = \frac{1}{N} (\alpha X^\alpha - (X - N)), \quad (9)$$

and they are both maximised at the same level of lending, X^{mon} .

Now, still for a given N , and assuming capital requirements are binding ($X = \frac{N}{\gamma}$), an increase in γ plays out in the same way as a decrease in X . So, the return on equity is also hump-shaped in γ and peaks at $\gamma^{\text{mon}} \equiv N/X^{\text{mon}}$. Hence, R_γ is positive when $\gamma < \gamma^{\text{mon}}$ and turns negative past that threshold. The intuition being that unless the level of lending is already lower than what a monopolist would choose (taking N as given), an increase in γ contracts credit and increases monopoly rents.

3.3.2. There is a threshold for γ that pins down the sign of the capital flows

Substituting $X = \frac{N}{\gamma}$ in Eq. (9) and adding back uncertainty, we return to the definition of the return on equity in our model (Eq. (4)):

$$R(N, \gamma) = \int_{A^0(N, \gamma)}^{A^H} \left(AN^{\alpha-1} \alpha \left(\frac{1}{\gamma} \right)^\alpha - \frac{1}{\gamma} + 1 \right) g(A) dA. \quad (10)$$

The presence of uncertainty does not affect the fact that restraining aggregate quantities can increase total revenue in the banking sector. The risk of bank failure implies that capital requirements are binding and banks specialise in equilibrium. From a normative perspective, it also motivates the use of capital requirements as policy tools. However, the introduction of uncertainty creates a technical challenge linked to the truncation in Eq. (10). To deal with this, we introduce the following condition.

Condition 1 (Regularity - Single Crossing). For all $\gamma' \in (0, 1)$, $R_{\gamma\gamma}(N, \gamma) < 0$ when $R_\gamma(N^*, \gamma) = 0$.

The fundamental drivers of a hump-shaped return on equity in our simple example is that: (i) the production function is concave, which generates a downward sloping and convex loan demand curve; (ii) capital requirements bind. The regularity condition ensures $R^*(\gamma, \gamma') \equiv R(N^*(\gamma, \gamma'), \gamma)$ is a well-behaved hump shape in γ . This allows us to streamline the analysis by ignoring knife-edge cases where $R_{\gamma\gamma}$ could be locally positive.¹¹

Theorem 1. Assume Regularity Condition 1 holds. Then, $\forall \gamma' \in (0, 1)$, there exists a unique $\hat{\gamma}(\gamma') > 0$ such that, $\forall \gamma \in (0, 1)$

$$\begin{cases} \frac{dN^*}{d\gamma} > 0 & ; \gamma < \hat{\gamma}(\gamma') \\ \frac{dN^*}{d\gamma} = 0 & ; \gamma = \hat{\gamma}(\gamma') \\ \frac{dN^*}{d\gamma} < 0 & ; \gamma > \hat{\gamma}(\gamma'). \end{cases}$$

The threshold $\hat{\gamma}(\gamma')$ can be interpreted as the capital requirement that maximises $R^*(\gamma, \gamma') \equiv R(N^*(\gamma, \gamma'), \gamma)$ given γ' . The broad intuition for the theorem is in line with that for the fictitious monopolist above. We convey it graphically in Fig. 1 (using again the simplification $A = 1$). The diminishing marginal return to physical capital translates into a downward sloping demand for loans (the blue line in the Figure). The supply of funds is depicted by the dashed black line. If equity capital is abundant (i.e. $\omega > N + N'$), the line is flat. The resulting equilibrium, with $z^* = 0$, corresponds to the black dot.

Raising γ will make capital scarce: M^* , and hence z^* , will become positive at some point. This amounts to an anticlockwise rotation in the supply curve. The quantity lent falls and the equilibrium shifts upwards along the blue demand curve. In line with the intuition for the monopolist above, the decline in the quantities first raises the return on equity up to a point and then decreases it. To see this, note that the bank's payout to shareholders (net of an opportunity cost of funds of 1) is $\alpha X^\alpha - X$: this quantity is the rectangle formed by the equilibrium point, the $z = 0$ supply curve and the y -axis. The net payout is maximised at the point $\gamma = \hat{\gamma}(\gamma')$, depicted by the red dot (and the area by the red rectangle). Maximising the net payout also maximises the return on equity, so R^* also peaks at $\gamma = \hat{\gamma}(\gamma')$.

¹¹ At $\frac{dR^*}{d\gamma} = 0$, the second derivative of $R^*(\gamma, \gamma')$ with respect to γ is $R_{\gamma\gamma}(N^*, \gamma)(1 - \xi^* R_N(N^*, \gamma))$. Hence, the second order condition for a maximum is satisfied given regularity Condition 1. Due to the truncation in $R(N, \gamma)$, $R_{\gamma\gamma} < 0$ is not always guaranteed. It can be positive if there is a local bump in the probability density function $g(A)$ in the direct vicinity of A_0 . Then, a small change in γ can affect the probability of default in a way that locally dominates the global force that makes the function hump-shaped in γ . Such issue is, e.g., ruled out if the equilibrium elasticity of the bank probability of survival with respect to the capital requirement is smaller than 1 (see the closed form results of Section 4 for an example).

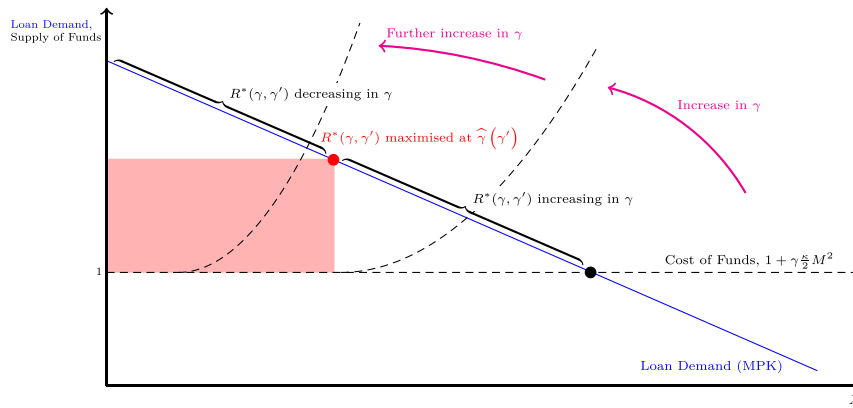


Fig. 1. Graphical Illustration of Theorem 1.

Notes - This illustration assumes no uncertainty ($A = 1$) and hence there is no distortion arising from deposit insurance. The blue line is the loan demand ($\alpha X^{\alpha-1}$). The black dashed lines are the supply of funds curve for given levels of the capital requirement γ , that is $1 + \gamma z$. Noting that $N = \gamma X$, the cost function is increasing and convex in X , unless capital is fully abundant (and $z = 0$). Furthermore, an increase in γ steepens the relationship between X and z . If a cost of funds curve is drawn for the corresponding $N^*(\gamma', \gamma)$, its intersection with the loan demand pins down the equilibrium. The red dot is then the point at which $R^*(\gamma, \gamma')$ is maximised, which implicitly defines $\hat{\gamma}(\gamma')$. The area of the red rectangle is given by $(\alpha X^{\alpha-1} - 1)X$, and we have seen in the simplified example above (also with $A = 1$), that maximising this object with respect to X also maximises return on equity when $A = 1$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

We prove the theorem for stochastic A . When banks are risky, the implicit subsidy from deposit insurance modifies the shape of the supply of funds curve in Fig. 1, but its presence does not affect the fundamental logic: a tighter capital requirement still rotates the supply curve anticlockwise and $R^*(\gamma, \gamma')$ is still hump shaped in γ .

In equilibrium banks cannot benefit from an increase in γ . $\hat{\gamma}(\gamma')$ is the value of the capital requirement that maximises the equilibrium return on equity. But, in any equilibrium, the return on equity is equated to the cost of raising equity (z^*): if a tighter requirement raises returns it must also raise costs. Investors are always indifferent at the margin.

Put differently, $\hat{\gamma}(\gamma')$ does not maximise bank pure-profits ($R^* - (1 + z^*)N^*$), simply because the latter includes the cost of bank equity capital. In our model, perfect competition implies that banks' pure profits are always zero in equilibrium, so a cost increase associated with a change in γ just offsets the increase in revenue. Although an increase in γ only makes the banks of our model *weakly* worse off, it would make them *strictly* worse off if they had market power (which is the case in reality). We show this in Appendix B.1.

4. The policy game

From now on, we treat γ and γ' as the choice variables of national regulators who take as given that, in equilibrium: (i) capital requirements are binding; and (ii) market forces will equate bank return on equity across countries (per Proposition 1).

4.1. Objective functions and solution concept

To start, our baseline assumption is that the excess cost of equity (z) is private and is not internalised by regulators. Whether bank capital is socially costly is a subject of debate in the banking literature (Admati et al., 2013). In our context, assuming costs are private helps with tractability and allows us to isolate the key externality that we wish to focus upon. However, as we will discuss, the presence of this externality and our main results do not hinge on this assumption.

We endow the Home regulator with the objective function

$$\begin{aligned} \pi^*(\gamma, \gamma') = & \underbrace{(X^*(\gamma, \gamma'))^\alpha - X^*(\gamma, \gamma')}_{\text{expected net output}} \\ & - \lambda \int_0^{A^0} \left(\underbrace{X^*(\gamma, \gamma') - N^*(\gamma, \gamma')}_{\text{liabilities}} - \underbrace{A\alpha (X^*(\gamma, \gamma'))^\alpha}_{\text{asset value}} \right) g(A) dA, \end{aligned} \quad (11)$$

where $\lambda > 0$ is a parameter.

With the specification in Eq. (11), we consider a trade off between economic activity and financial stability. The first term, $(X^*)^\alpha - X^*$, captures the expected economic surplus generated by Home banks' lending (recall $\mathbb{E}[A] = 1$). To model the financial stability motive, we impose a deadweight cost of bank defaults in the second term that is proportional to the expected shortfall between bank liabilities and its assets in the event of default. In more compact notation, which will be useful to express partial derivatives, this shortfall reads:

$$S(X, N) \equiv \int_0^{A^0} \left(\underbrace{X - N}_{\text{liabilities}} - \underbrace{A\alpha X^\alpha}_{\text{asset value}} \right) g(A) dA.$$

The mechanisms we study here do not hinge on the specific form of $S(X, N)$. As we shall see, its key features lie in the sign of its partial derivatives: $S_X(X, N) \geq 0$ and $S_N(X, N) \leq 0$, which we argue are quite natural.

First, $S_X(X, N) \geq 0$: at a given N , an increase in X increases social costs of bank insolvency. This is natural for two reasons: (i) for a given N , more lending means more leverage; and, ii) lending itself is likely to exhibit diminishing marginal returns. Hence, average asset quality goes down too. Second, $S_N(X, N) \leq 0$: At a given X , more bank equity capital decreases deadweight losses. So, unless banks are completely safe, bank equity capital is valuable because it decreases the costs of insolvency.

Expressing the cost of financial instability in such a fashion has become popular since the global financial crisis. As is discussed in Malherbe (2020), such costs can be seen as stemming from the resolution process, whose complexity is likely to increase in the shortfall (as is for instance the case in the costly verification set up of Townsend (1979)). Other possible micro-foundations include the tax distortions of associated bailouts (see, e.g., Acharya et al. (2010)): given our tie breaking rule for incorporation, the shortfall is the deposit insurance bill at Home and so corresponds to the direct fiscal cost of bank failures. More generally, the specification could capture any other channel by which bank financial distress affects the real economy (Hoggarth et al., 2002). From this point of view, our approach to the cost of bank financial distress is also similar to many papers in the systemic risk literature.¹²

¹² Given we have a representative bank, our expression for $S(X, N)$ is equivalent to the systemic expected shortfall in Acharya et al. (2017), up

The partial derivative, with respect to X , of the right side of (11), defines the social net present value of the marginal loan:

$$\text{NPV} \equiv \alpha(X)^{\alpha-1} - 1 - \lambda S_X.$$

The intrinsic net present value from lending is $\alpha(X)^{\alpha-1} - 1$: the marginal product of capital net of an opportunity cost of funds of 1. The social NPV also accounts for the deadweight losses generated by the marginal loan, λS_X . Banks do not internalise deadweight losses, and they benefit from the implicit subsidy from deposit insurance. This is why the market equilibrium generally does not implement $\text{NPV}^* = 0$, and constitutes a rationale for capital regulation.

Solution concepts. We consider both competition and collaboration between regulators. When they collaborate, the regulators jointly set $\gamma = \gamma' = \gamma^{\text{col}}$ to maximise their combined objective such that

$$\gamma^{\text{col}} \equiv \arg \max_{\gamma=\gamma'} \pi^*(\gamma, \gamma') + \pi^{*'}(\gamma', \gamma). \quad (12)$$

When two regulators compete, we consider a symmetric Nash equilibrium of the game in which they separately maximise $\pi^*(\gamma, \gamma')$ and $\pi^{*'}(\gamma', \gamma)$, setting their capital requirement taking the other's as given. Formally, a Nash equilibrium is defined as a fixed point

$$\gamma^{\text{nash}} = \arg \max_{\gamma} \pi^*(\gamma, \gamma^{\text{nash}}).$$

Henceforth, functions with superscripts ^{nash} are evaluated at the symmetric Nash equilibrium of the competitive regulator game. For example, $N^{\text{nash}} \equiv N^*(\gamma^{\text{nash}}, \gamma^{\text{nash}})$. Likewise, functions with superscripts ^{col} are evaluated at the market equilibrium where both capital requirements are set at γ^{col} .

Condition 2 (Regularity - Single Crossing). For all $\forall \gamma, \gamma' \in [0, 1]$: $\pi^*(0, \gamma') > 0$, $\pi^*(1, \gamma') < 0$, $\pi^*_{\gamma\gamma'}(\gamma, \gamma') < 0$, $\pi^*_{\gamma\gamma'}(\gamma, \gamma') < 0$, and the same holds for $\pi^*(\gamma, \gamma') + \pi^{*'}(\gamma', \gamma)$.

Condition 2 ensures that the first order condition to problem (12) uniquely pins down γ^{col} . It also ensures the individual competitive regulators' first order conditions define well behaved best response function leading to a unique γ^{nash} .¹³

4.2. Race to the top or the bottom?

The main question we are now interested in is whether Nash competition leads to tighter or looser requirements than collaboration. As we will see, the capital flow described in Section 3 has a key role to play in determining how competition plays out versus collaboration.

Nash inefficiency: the role of the capital flow externality. The first order condition for γ^{col} reads (see the proof of Proposition 3)

$$\pi^*_{\gamma}(\gamma, \gamma') + \underbrace{\left(\frac{\text{NPV}^*}{\gamma'} - \lambda S_{N'}^* \right) \frac{dN^*}{d\gamma}}_{=\pi^{*'}_{\gamma}(\gamma', \gamma)} = 0. \quad (13)$$

to a shifting constant. Other examples of approaches based on the shortfall include (Lehar, 2005), Brownlees and Engle (2017), and Indergand et al. (2023). Additionally, see Duarte and Eisenbach (2021) for the empirical association between fire sales exposure and such measures.

¹³ We impose this restriction for the same reason as Condition 1. The truncation at A_0 means the objective function in (11) may have multiple peaks, or exhibit jumps or kinks. In these knife-edge cases, there may, for instance, be either no or several γ' s that solve the first order conditions. This can occur if there is a large probability mass at A_0 . Ignoring these cases enables a first order approach (Bahaj and Malherbe, 2020): focusing on small increases in γ helps build intuition and derive formal results in our propositions. Fundamentally, the condition is a restriction on $g(A)$, which we had kept general so far. Below we derive closed form results when we restrict A to a binary distribution, which ensures Condition 2 is satisfied.

Let us focus on the second term. It captures the effect on Foreign of a marginal increase in γ . First, consider the case in which a change in γ attracts equity capital to Home. If $\frac{dN^*}{d\gamma} > 0$, it must be that some of this capital has been siphoned off from Foreign: $\frac{dN^{*'}}{d\gamma} < 0$ (Proposition 2).¹⁴ Equity capital is valuable to Foreign. It can be levered up at rate $1/\gamma'$ to fund lending with a marginal social value $\text{NPV}^{*'} > 0$.¹⁵ Equity capital also decreases the expected shortfall on Foreign bank assets, that is $S_{N'}^{*'} < 0$. So the term in parenthesis is positive and the loss of capital hurts Foreign.

Hence, changes in γ generate an externality that operates through capital flows. And, in the case where $\frac{dN^*}{d\gamma} > 0$, this externality is negative: Foreign welfare is falling in the Home requirement: $\pi^{*'}_{\gamma}(\gamma', \gamma) < 0$. Now, if Eq. (13) holds and $\pi^{*'}_{\gamma}(\gamma', \gamma) < 0$, then it must be that $\pi^*_{\gamma}(\gamma, \gamma') > 0$. But this precisely means that the competitive Home regulator has an incentive to deviate upwards at the collaborative optimum. Part of the cost of tighter Home requirements is borne abroad and so regulatory competition leads to overly tight regulation: as we will see, when $\frac{dN^{\text{col}}}{d\gamma} > 0$, we get $\gamma^{\text{nash}} > \gamma^{\text{col}}$, which we interpret as a race to the top.

However, when $\frac{dN^*}{d\gamma} < 0$ the sign of the capital flow externality flips. We have $\pi^{*'}_{\gamma}(\gamma', \gamma) > 0$ because a higher home requirement then leads to an inflow of equity capital to Foreign. More capital boosts lending and reduces the expected shortfall, so tighter Home regulation acts as a positive externality. In this case, at the collaborative optimum, the positive externality, $\pi^{*'}_{\gamma}(\gamma', \gamma) > 0$, implies $\pi^*_{\gamma}(\gamma, \gamma') < 0$. The competitive Home regulator has an incentive to deviate towards looser regulation: when $\frac{dN^{\text{col}}}{d\gamma} < 0$, we get $\gamma^{\text{nash}} < \gamma^{\text{col}}$, which we interpret as a race to the bottom.

Whether competition leads to a race to the top or the bottom depends on the sign of $\frac{dN^{\text{col}}}{d\gamma}$: the direction of the capital flow following a unilateral deviation towards tighter regulation (starting from collaboration). From Theorem 1, we know that the sign of $\frac{dN^{\text{col}}}{d\gamma}$ depends on whether γ^{col} is greater than or less than the relevant return maximising requirement.

We now state this result formally:

Proposition 3. Assume Conditions 1 and 2 hold. Then, there are three cases

$$\begin{cases} \gamma^{\text{col}} < \gamma^{\text{nash}} < \bar{\gamma} & \text{and } \frac{dN^{\text{col}}}{d\gamma} > 0 \\ \gamma^{\text{col}} = \gamma^{\text{nash}} = \bar{\gamma} & \text{and } \frac{dN^{\text{col}}}{d\gamma} = 0 \\ \gamma^{\text{col}} > \gamma^{\text{nash}} > \bar{\gamma} & \text{and } \frac{dN^{\text{col}}}{d\gamma} < 0, \end{cases}$$

where $\bar{\gamma}$ is the fixed point $\bar{\gamma} = \hat{\gamma}(\bar{\gamma})$, with $\hat{\gamma}(\cdot)$ defined in Theorem 1.

The broad intuition for the proposition is straightforward. The relation between γ^{nash} and γ^{col} in the three cases follows the reasoning above: the incentive to deviate from collaboration depends on the sign of the capital flow. If the collaborative requirement is less than the return maximising one ($\gamma^{\text{col}} < \bar{\gamma}$) a unilateral deviation from collaboration towards tighter regulation is optimal as a tighter regulation generates capital inflows ($\frac{dN^{\text{col}}}{d\gamma} > 0$). So $\gamma^{\text{col}} < \bar{\gamma} \Leftrightarrow \gamma^{\text{col}} < \gamma^{\text{nash}}$ and vice versa.

That $\bar{\gamma}$ places a bound on how far γ^{nash} can deviate from γ^{col} is perhaps less important but it is also an intuitive result. Take the case

¹⁴ If $\omega \geq 2\alpha \frac{1}{1-\alpha}$, equity capital is plentiful enough to finance all positive NPV loans solely with equity. So we would have $\gamma^{\text{col}} = 1$: under those circumstances $\text{NPV}^* = 0$ and $S_{N'}^* = 0$ as banks never default. The restriction $\omega < 2\alpha \frac{1}{1-\alpha}$ ensures that this is not the case and that there are spillovers at the collaborative outcome (i.e., $\text{SP}^{\text{col}} > 0$ and $\frac{\text{NPV}^*}{\gamma'} - \lambda S_{N'}^* \neq 0$).

¹⁵ At the collaborative optimum, NPV^{*} is positive (this is not guaranteed out of equilibrium). If it were negative, regulators could costlessly increase capital requirements. This would reduce negative NPV lending, raise the social NPV on the marginal loan and decrease deadweight losses. All of which are beneficial.

$\gamma^{\text{col}} < \bar{\gamma}$ and imagine that (out of equilibrium) both regulators picked $\bar{\gamma}$. By definition of $\bar{\gamma}$, there is no capital flow externality from a marginal deviation. So, individual and collective incentives are the same. But we know that $\gamma^{\text{col}} < \bar{\gamma}$, so the beneficial deviation should be towards a γ strictly lower than $\bar{\gamma}$. So we have $\gamma^{\text{nash}} < \bar{\gamma}$. Again, the symmetric reasoning applies to the other case ($\gamma^{\text{col}} > \bar{\gamma}$). Finally, that $\gamma^{\text{col}} = \gamma^{\text{nash}}$ when $\gamma^{\text{col}} = \bar{\gamma}$ is clear as the externality is nil at the collaborative optimum so there is no incentive to deviate.

Now, whether $\bar{\gamma}$ is greater or smaller than γ^{col} depends on the primitives of the model: parameters $\alpha, \kappa, \omega, \lambda$ and the distribution $g(A)$. Making specific assumptions on some of them allows to derive intuition on the role of the others. This will be our strategy for characterising the Nash equilibrium further.

Remark. With our specified objective function (11), the capital flow externality emerges, in part, because a reallocation of equity capital between countries affects the expected shortfall from bank defaults. However, regulatory objectives could be modelled in a variety of ways. We show in Appendix B.2 that objective (11) is a specific case of a more general class of objective function that still results in Proposition 3. In particular, for the Proposition to hold, it is sufficient to have an objective where (i) Foreign capital requirement only matters for Home welfare through the equity capital of Home banks, $N^*(\gamma, \gamma')$, and (ii) at the collaborative optimum, given X^* , an additional unit of equity financing is welfare enhancing.¹⁶

The second condition in the remark can be interpreted as the regulators finding bank equity capital socially valuable. Meaning that an inflow of equity capital from abroad would lead to an improvement in welfare at home, whereas siphoning capital off one's neighbour imposes a negative externality. In the case of our specific objective, more equity capital would enable banks to lend the same amount but reduce the extent of defaults, which is welfare enhancing. The policy debate is often pitched in these terms so this condition seems to correspond to the empirically relevant case. However, we discuss a case where the social value of capital can flip sign in Section 5.3.

The notion that Foreign capital regulation only matters for Home welfare through the allocation of equity capital abstracts from other potential sources of spillovers between countries from regulation. We show in Appendix B.2, that the capital flow externality identified in this section still exists and has a similar interpretation even if there are other sources of spillovers (e.g., if z^* is perceived as socially costly).

4.3. Characterising the Nash equilibrium: the role of parameters

We now turn to question of whether or not $\gamma^{\text{col}} > \bar{\gamma}$ and explore the role of the model's primitives. To do so, we assume $\kappa \rightarrow \infty$. This simplifies expressions and allows for sharper intuition from analytical results, some of which are in closed form.

With $\kappa \rightarrow \infty$, any allocation with $\gamma = \gamma'$ will result in $N^* = N'^* = \omega/2$. Given capital requirements are binding, we have $X^{\text{col}} = \frac{\omega}{2\gamma^{\text{col}}}$ and we can express the collaborative maximisation problem in terms of X rather than γ . Hence, the collaborative first order condition is

$$\underbrace{\alpha (X^{\text{col}})^{\alpha-1}}_{\text{MPK}} = 1 + \lambda \underbrace{\int_0^{A^0} (1 - \alpha^2 A (X^{\text{col}})^{\alpha-1}) g(A) dA}_{\equiv S_X^{\text{col}} \geq 0}. \quad (14)$$

¹⁶ To see this formally, note the first condition in the remark implies a welfare function of the following form

$$\pi^*(\gamma, \gamma') = V(N^*(\gamma, \gamma'), \gamma).$$

Then the externality on Foreign following a Home requirement increase is $\pi'_{\gamma'}(\gamma', \gamma) = V'_{N'}(N'^*(\gamma', \gamma), \gamma) \frac{dN'^*}{d\gamma}$. The second condition in the remark ensures $V'_{N'}(N'^*(\gamma^{\text{col}}, \gamma^{\text{col}}), \gamma^{\text{col}})$ is positive. So the direction of a profitable deviation at the collaborative optimum is given by the sign of $\frac{dN'^*}{d\gamma}$. This leads to the same economic intuition discussed when the objective is given by Eq. (11).

This condition reflects that collaborative regulators equate the marginal product of physical capital (MPK) to one plus a positive wedge (λS_X), this is equivalent to setting $\text{NPV}^* = 0$. The reason why collaborative regulators only focus on NPV^* in this case (and ignore S_N^*) is that they cannot affect the equilibrium global allocation of equity capital: $N^* = N'^* = \omega/2$. (Unilateral deviations, however, will still affect its cross border allocation.)

From Section 3.2, the symmetric return-maximising capital requirement $\bar{\gamma}$ solves $R_{\gamma}(\omega/2, \bar{\gamma}) = 0$. From this, we can define the return-maximising level of lending $\bar{X} \equiv \frac{\omega/2}{\bar{\gamma}}$. It solves

$$\alpha \times \underbrace{(\bar{X}^{\alpha-1})}_{\text{MPK}} + \int_0^{A^0} (1 - \alpha^2 A \bar{X}^{\alpha-1}) g(A) dA = 1. \quad (15)$$

We will come back to the interpretation of condition (15) in a few lines. For now, it is important to recognise: $\bar{\gamma} \leq \gamma^{\text{col}} \Leftrightarrow \bar{X} \geq X^{\text{col}}$. Whether X^{col} is greater than or less than $\bar{X}$ will tell us which of the three cases from Proposition 3 arises.

The role of ω . Let us start with highlighting the role of ω .

Proposition 4. Assume $\kappa \rightarrow \infty$, Conditions 1 and 2 hold, and $g(A)$ is continuous for all $A^0 > 0$, with full support over $[0, A^H]$. Then $\forall \alpha, \lambda, \exists \omega \leq \bar{\omega} < 2\alpha^{\frac{1}{1-\alpha}}$:

$$\begin{cases} \gamma^{\text{col}} < \gamma^{\text{nash}} < \bar{\gamma} & ; \omega > \bar{\omega} \\ \gamma^{\text{col}} = \gamma^{\text{nash}} = \bar{\gamma} & ; \omega = \bar{\omega} \\ \gamma^{\text{col}} > \gamma^{\text{nash}} > \bar{\gamma} & ; \omega < \bar{\omega}. \end{cases}$$

Broadly speaking, Proposition 4 emerges because X^{col} is weakly increasing in ω , whereas $\bar{X}$ is weakly decreasing. This means there is a threshold value $\bar{\omega}$ where the two cross and the competitive outcome switches between a race to the top and race to the bottom.

To understand why X^{col} is weakly increasing in ω , first note that $A^0 = \frac{X - \omega/2}{\alpha(X)^{\alpha}}$ is decreasing in ω . Hence, larger equity capital buffer decreases the likelihood of default. So, S_X^{col} decreases with ω : at the margin, deadweight losses are lower with a higher ω . This implies that X^{col} is increasing in ω . Furthermore, as equity capital becomes large, the marginal deadweight loss of bank default vanishes. That is, for ω large enough, $A^0 \rightarrow 0$ and $S_X^{\text{col}} \rightarrow 0$. Then, private and social NPV coincide and X^{col} simply maximises the economic surplus from production: $\alpha (X^{\text{col}})^{\alpha-1} \rightarrow 1$.¹⁷

Now, to understand why $\bar{X}$ is weakly decreasing in ω , consider condition (15). The second term, in the integral, is a positive wedge that reflects the implicit subsidy arising from deposit insurance. This wedge effectively raises the perceived private marginal return on lending. Return maximisation takes advantage of banks' ability to risk shift onto the taxpayer by taking on marginal loans with relatively low returns given that the bank does not bear the associated downside (Kareken and Wallace, 1978). As we explained above, when ω increases, A^0 decreases. This reduces the wedge, which leads to a reduction in $\bar{X}$.

Ultimately, as $A^0 \rightarrow 0$, we get¹⁸

$$\alpha (\bar{X})^{\alpha-1} \rightarrow \frac{1}{\alpha}. \quad (16)$$

This limit case exactly corresponds to that of our fictitious monopolist of Section 3.2, where $\frac{1}{\alpha}$ reflects the markup associated with monopolist

¹⁷ Note that if $\frac{\omega}{2} > \alpha^{\frac{1}{1-\alpha}}$ there would be more than enough equity capital to fully finance such $X^{\text{col}} = \alpha^{\frac{1}{1-\alpha}}$. So we would have $\gamma^{\text{col}} = 1$ instead of $\gamma^{\text{col}} = \frac{\omega/2}{X^{\text{col}}}$. In addition, we would also have $\gamma^{\text{nash}} = 1$. As already noted, if equity capital were fully abundant, there would be no capital flow externality. We restricted the parameter space to $\omega < 2\alpha^{\frac{1}{1-\alpha}}$ to rule this out.

¹⁸ Note that if $\frac{\omega}{2} > \alpha^{\frac{1}{1-\alpha}}$ there is also more than enough equity capital to fully finance such $\bar{X}$. In that case, we have $\bar{\gamma} > 1$. For all $\omega < 2\alpha^{\frac{1}{1-\alpha}}$, $\gamma^{\text{col}} < 1$, so $\gamma^{\text{col}} < \bar{\gamma}$. This is why $\bar{\omega}$ must be strictly below $2\alpha^{\frac{1}{1-\alpha}}$.

marginalisation (this also explains why MPK is marginalised by an additional factor, α , in condition (15)).

Given this, it is clear that, when ω is large enough $X^{\text{col}} > \bar{X}$, which is equivalent to $\gamma^{\text{col}} < \bar{\gamma}$. And, in such a case, regulators have an incentive to deviate upwards from γ^{col} and we get a race to the top ($\gamma^{\text{col}} < \gamma^{\text{nash}}$, as per Proposition 4). As ω decreases, $\bar{X}$ increases and X^{col} decreases. So, it is more likely that $X^{\text{col}} < \bar{X}$ for relatively low ω . Depending on the value of the other parameters, it can be the case that even at $\omega \rightarrow 0$, we still have $X^{\text{col}} > \bar{X}$. Then, only the race to the top is possible ($\bar{\omega} = 0$ in this case).¹⁹

4.3.1. A closed form case to isolate the role of other parameters

To delve into the role of the other parameters, we now neutralise the marginal effect of ω on A_0 by assuming that $g(A)$ is binary: $A \in \{0, \frac{1}{q}\}$, with $\Pr\left(\frac{1}{q}\right) = q$. Then the key conditions (14) and (15) reduce to

$$\begin{aligned} \alpha (X^{\text{col}})^{\alpha-1} &= 1 + \lambda(1-q) \quad (\text{Collaborative FOC}) \\ \frac{1}{q} \times \alpha (\bar{X})^{\alpha-1} &= \frac{1}{\alpha} \quad (\text{Return maximising condition}) \end{aligned} \quad (17)$$

which gives us closed-form solutions for both X^{col} and $\bar{X}$ and, in turn, for the corresponding capital requirements:

$$\bar{\gamma} = \frac{\omega}{2} \left(\frac{q}{\alpha^2} \right)^{\frac{1}{\alpha-1}} \quad \text{and} \quad \gamma^{\text{col}} = \frac{\omega}{2} \left(\frac{1 + \lambda(1-q)}{\alpha} \right)^{\frac{1}{\alpha-1}}.$$

It directly follows that:

$$\gamma^{\text{col}} \leq \gamma^{\text{nash}} \Leftrightarrow \lambda \leq \frac{q - \alpha}{(1-q)\alpha}.$$

We can now run through the intuition for the role played by other parameters.

The role of λ . The higher λ , the higher are deadweight losses at the margin. Hence, a higher λ will lead regulators to choose a more constrained level of equilibrium lending, which translates to a higher γ^{col} . Since λ does not affect $\bar{X}$, the higher λ the more likely $\bar{X} > X^{\text{col}}$, and therefore $\gamma^{\text{nash}} < \gamma^{\text{col}}$.

The role of bank riskiness. When discussing the role of ω , we stressed its impact on A^0 . If bank equity capital is less scarce, the probability of bank default goes down. This had two effects: (i) a reduction in expected deadweight losses; and, (ii) a decrease in the implicit subsidy from deposit insurance. The two effects are more transparent in the binary distribution case. The first materialises with the factor $1 - q$ in the Collaborative FOC, and the second with the factor $\frac{1}{q}$ in the revenue maximising condition. The latter reflects that, with deposit insurance and limited liability, the return on equity only depends on the upside. The realised return on equity is always nil when the bank goes bust. Therefore, the value for X that maximises the return on equity only depends on the value of A when the bank survives (rather than its expectation): namely $1/q$. Holding $\mathbb{E}[A] = 1$, the better the upside (i.e., the lower q) the higher $\bar{X}$. Together, it is clear that the lower q the more likely $\bar{X} > X^{\text{col}}$, and therefore $\gamma^{\text{nash}} < \gamma^{\text{col}}$. This reinforces our stressing of the role of ω .

The role of α . We have established in the continuous $g(A)$ case that, as ω grows large and banks become safe, $\bar{X} < X^{\text{col}}$ because banking sector return maximisation implies a markup due to monopolist marginalisation. This feature extends to the case with risky banks and is reflected by the term $1/\alpha$ on the right side of Eq. (17). The smaller α the less elastic the loan demand, the larger the markup, and the more likely $\bar{X} < X^{\text{col}}$, and therefore $\gamma^{\text{col}} < \gamma^{\text{nash}}$.

¹⁹ This will for instance happen if λ is relatively small. This is because, then, deadweight losses do not matter much, and there is no reason for the regulators to constrain lending to levels so low that the capital flow becomes negative. But if λ is large enough, then it can be that, at some $\bar{\omega} > 0$, X^{col} just equates $\bar{X}$. At such a point, the capital flow is nil, and so is the externality. Hence $\gamma^{\text{col}} = \gamma^{\text{nash}} = \bar{\gamma}$. At even lower levels of ω , we then get $\gamma^{\text{col}} > \bar{\gamma}$, the externality is reversed, and we get a race to the bottom ($\gamma^{\text{col}} < \gamma^{\text{nash}}$).

4.4. The case with more than two countries

Our policy game can be extended to more than two countries. Take the case $\kappa \rightarrow \infty$ and consider a number of ex-ante identical countries, each with an initial investor endowment of $\omega/2$. The collaborative optimum is then the same as in the corresponding two-country case (i.e., γ^{col} and N^{col} would be unchanged). In addition, the direction in which regulators have an incentive to deviate would still depend on whether increases in capital requirement raise the return on equity at the collaborative optimum. However, the number of countries affects γ^{nash} and, in particular, its distance from γ^{col} .

To save on notation we proceed numerically: the left panel in Fig. 3 presents an illustration. For the two-country case, it displays γ^{col} , $\bar{\gamma}(\gamma^{\text{col}})$ and γ^{nash} , which is at the intersection of the best response curves. In addition, it presents the solution for γ^{nash} when, rather than two countries, there is a mass of atomistic countries. The example we provide is one where $\gamma^{\text{col}} < \gamma^{\text{nash}}$.

The more countries the greater the incentive to deviate and the further the Nash equilibrium from the collaborative optimum (as indicated by the atomistic case in Panel A of Fig. 3). The reason is that countries become effectively “smaller” on a global scale and, therefore, face a more elastic inverse supply curve of bank capital. Take the case where an increase in γ in a country siphons off capital from abroad. Capital flows will ensue so as to restore the no-arbitrage condition in international capital markets. As the capital outflow is spread across more and more countries, the total inflow into the deviating country (for an identical deviation) becomes greater. As a result, the deviating country can achieve an identical financial stability benefit at the cost of a lower and lower decrease in economic activity.

From a collaborative point of view, each country attempts to *overuse* equity capital in the Nash equilibrium. Not unlike in the tragedy of the commons, this gets worse as the number of users becomes large. Such an interpretation applies to the case where $\gamma^{\text{col}} < \gamma^{\text{nash}}$, which we just discussed and can be interpreted as a race to the top. But it also applies to $\gamma^{\text{nash}} < \gamma^{\text{col}}$. In this case, competition among regulators leads to a race to the bottom instead. However, this occurs precisely because cutting capital requirements attracts capital from abroad.

5. Discussion and extensions

Here we discuss how our host-country rule relates to real world capital regulation under the Basel III Accords. We then turn to policy implications and empirical predictions, and draw links with the stylised facts and existing empirical literature. Last, we consider an extension of the model in which deposits offer households a liquidity service, which allows us to discuss further the robustness of our analysis.

5.1. Real world capital requirements

The current Basel III regime that coordinates real world capital regulation across countries can be approximated as a static common minimum requirement (made up of a variety of different “buffers”) plus a time varying macroprudential requirement called the countercyclical capital buffer (CCyB). Hence, we can model a Basel III capital requirements for a bank i (at time t) as follows:

$$n_{it} \geq (\underline{\gamma} + \text{CCyB}_t)x_{it} + (\underline{\gamma} + \text{CCyB}'_t)x'_{it} \quad (18)$$

where $\underline{\gamma}$ is the fixed common minimum, and $\text{CCyB}_t, \text{CCyB}'_t \geq 0$ are the time-varying countercyclical buffers. Under Basel III rules, the CCyB is, in effect, set by the regulator of the borrowers' country. This is known as the principle of reciprocity in the Basel Accords. Regardless of where bank i is located it faces the CCyB_t on Home loans (and CCyB'_t on Foreign loans).

In practice, banks have three ways to lend across borders: directly, through branches, or through subsidiaries incorporated in the country of the borrower. Direct lending and lending made through branches fall

under the jurisdiction of the country where the bank is incorporated and so, *de jure*, the capital requirement on those loans is imposed by the bank's local regulator. However, the Accords mandate that countries impose, on their banks, for their foreign lending, the CCyB chosen by the corresponding host country (and the host country does the same; hence reciprocity). So, *de facto*, branches, subsidiaries, and direct cross-border loans all face the same capital requirement set by the country where the lending takes place. This is what motivates the host-country rule of our model.

Three remarks are in order. First, the CCyB cannot be set at a negative level. This is something we abstract from in our model but would be straightforward to incorporate with a constraint $\gamma, \gamma' \geq \underline{\gamma}$. Second, the Accords also stipulate a limit of 2.5% for mandatory reciprocation of the CCyB (i.e. the legal obligation to reciprocate stops when the buffer reaches 2.5%). Voluntary reciprocation is still possible and, in reality, is seen for other relevant macroprudential tools. However, no country has set the CCyB past 2.5% so far. So, the willingness to voluntarily reciprocate larger buffers cannot be judged. Third, reciprocity requires that the local regulator actually adheres to the principle. Publicly refusing to implement a policy legally mandated under the Basel accords is unlikely. A potential concern is that supervisors could be secretly lax on imposing the additional reciprocated capital requirement. But this would come with reputational and legal risks and may lead to the jurisdiction being deemed Basel non-compliant, leading to barriers for its banks international operations.²⁰ Such a strategy would also require collusion with the bank itself. An observed lack of integrity by supervisors may hinder other interactions with the banking system with consequences for wider financial stability. (For more detail on the institutional background, see [Appendix A](#).)

5.2. Empirical predictions and policy implications

A formal empirical analysis of the CCyB regime is beyond the scope of this paper. The short time frame since the CCyB were introduced and the relatively infrequent changes in the CCyB are significant hurdles. Still, we highlight how our model may shed light on the use of such a policy tool, and we then argue that one can also draw some inference on the sign of the capital flows based on existing literature.

To set the stage, consider the following thought experiment. Imagine a world in which a race-to-the-bottom is paramount but there is a common minimum to mitigate it (in our framework, $\gamma^{\text{nash}} < \underline{\gamma}$). If countries are symmetric, CCyBs would always be nil. But CCyBs are used in practice (see [Fig. 3](#), panel B). That a race to the top can arise in our model provides a potential explanation

Other mechanisms can lead to a race to the top.²¹ And other reasons why CCyBs are used may of course exist, especially if countries are heterogeneous. However, our analysis leads to novel predictions. We discuss them now.

5.2.1. Implication 1: CCyBs are likely to overshoot

In normal times, when the banking sector is healthy and bank capital is relatively abundant (ω is relatively high), we are more likely to be in a situation where national regulators have an incentive to deviate upward ($\gamma^{\text{nash}} > \gamma^{\text{col}}$). Doing so would attract capital from abroad and impose a negative externality onto other countries; this follows from [Proposition 4](#). So national regulators have an incentive

to tighten the CCyB too much in good times.²² Hence, there are gains from coordinating on smaller raises. Conversely, after a negative shock, when ω is low (and λ could be higher too, for instance if deadweight losses emanate from tax distortions and fiscal capacity has been impaired by the shock), regulators have incentives to cut the requirement too aggressively. Hence, there are gains from coordinating on smaller cuts.

Consider a version of our model where the single period would be repeated over time, with exogenous changes in ω . Our model predicts that γ^{col} would rise with ω (in line with conventional wisdom). However, the lower ω the more likely $\gamma^{\text{nash}} < \gamma^{\text{col}}$. So, the prediction is that changes in γ^{nash} will follow changes in γ^{col} but with a greater amplitude. We illustrate this in [Fig. 2](#).

As described, the Basel accords also specify limits to the CCyB. Interestingly, through the lens of our model, these limits could mitigate strategic behaviour. In bad times, if we are in the case where $\gamma^{\text{nash}} < \underline{\gamma}$, we would have bunching at $\underline{\gamma}$ as the positivity constraint on the CCyB limits the scope for regulators to undercut one another. Whereas, in good times, if a national regulator sets the buffer above 2.5%, other regulators will not be obliged to reciprocate.

5.2.2. Implication 2: Small countries are likely to use CCyBs more

An implication of our model with more than two countries is that smaller countries should be most active in using the CCyB.²³ These countries have the greatest incentive to deviate upwards during good times and to cut during bad times as, from their perspective, the supply of bank equity capital is more elastic. While other forces may be at work, this is what we have seen in reality. The most active users of the CCyB have been small economies in Scandinavia, Hong Kong and the smaller members of the Eurozone. At the time of writing, the U.S., China and Japan have not used the buffer while Germany has varied the buffer by 25bp since its introduction. To illustrate this point more concretely in the data, the right panel of [Fig. 3](#) plots the May 2023 level of the CCyB across the 28 members of the Basel Committee (and Norway) against the country's share in the total of members GDP (log scale). 2023 represents a period where many countries are increasing buffers after having cut them during the pandemic and so we would expect country size to be negatively correlated with CCyB rates. That is what we see in the data. The evidence is purely suggestive, but it is worth flagging that country size alone accounts for 16% of the cross-sectional variation in CCyB rates.

5.2.3. Is there empirical evidence for the sign of the capital flow?

There is a substantial literature identifying the effect of capital requirements at the bank level by exploiting the heterogeneous impact of regulatory reforms, stress tests, or supervisory interventions ([Francis and Osborne, 2012](#); [Bahaj et al., 2016](#); [Gropp et al., 2019](#); [Imbierowicz et al., 2018](#); [Juelsrud and Wold, 2020](#); [De Jonghe et al., 2020](#)). A message from these papers is that banks facing an increase in capital requirement increase their equity.²⁴ In some studies, the response is not significantly different than zero, but we are not aware of evidence that banks, on average, reduce their levels of capital in response to higher requirements.

In our model, this bank-level relationship corresponds to $\frac{dn}{d\gamma} \geq 0$. Extrapolating at the aggregate level, this suggests $\frac{dN}{d\gamma} \geq 0$ and,

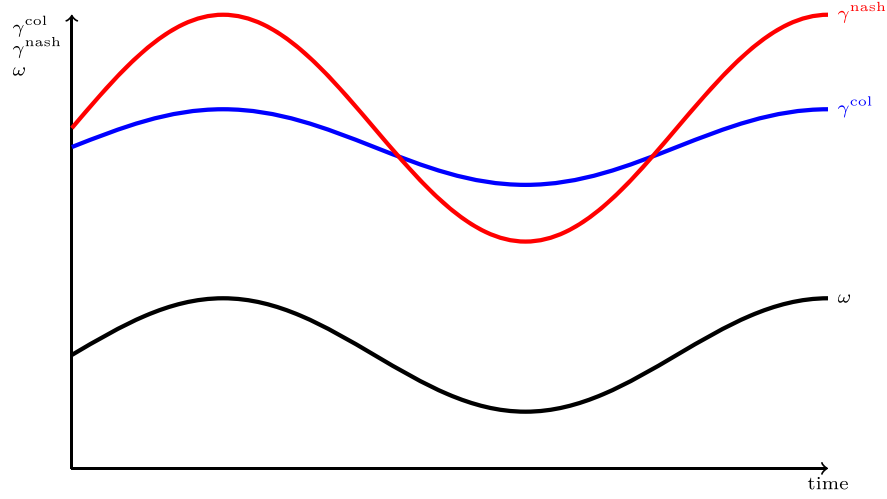
²⁰ The Basel Committee has a Regulatory Consistency Assessment Programme to judge members adherence to the common rulebook and to identify deviations.

²¹ For example, this could be due to a terms of trade externality (as in [Haufler and Maier \(2019\)](#)) or because of spillovers from bail out costs ([Gersbach et al., 2020](#)).

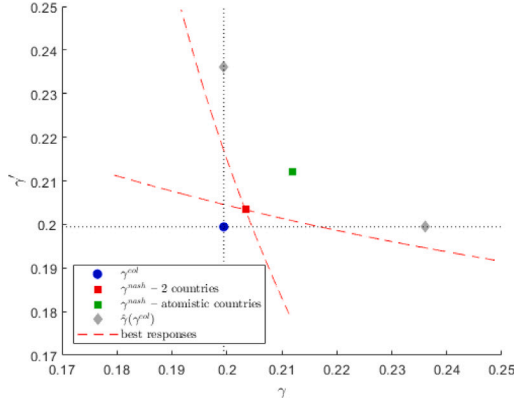
²² There seems to be a consensus in the literature and among policymakers that capital requirement should be set higher in good times than in bad times. A key reason is that bank capital is perceived to be scarcer in bad times (see, e.g., [Kashyap and Stein \(2004\)](#), and [Malherbe \(2020\)](#)).

²³ We have not formally considered asymmetries in country size (or along other dimensions) within our framework. Differences in country size can generate interesting predictions for the coordination of policies in other contexts (e.g. see [Opp \(2010\)](#) for an analysis of tariffs). We postpone such an analysis to future research.

²⁴ See also [Bahaj and Malherbe \(2020\)](#) for a theoretical analysis.

Fig. 2. Illustration of $\gamma^{nash} \leq \gamma^{col}$ over the cycle.

Panel A: Equilibrium outcomes



Panel B: Country size versus CCyB rates in 2023

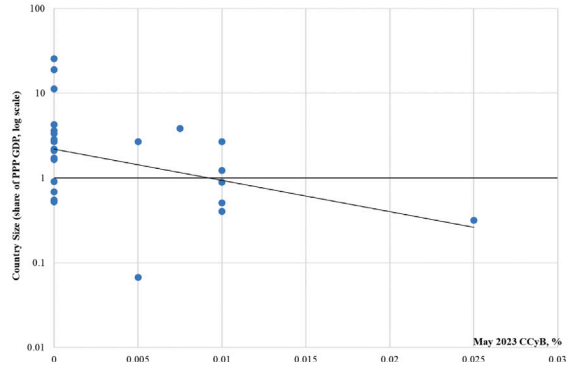


Fig. 3. Case with more than 2 countries.

Notes - Parameter choices: $\log(A) \sim N(0, \sigma^2)$, $\sigma^2 = 0.02$, $\lambda = 5$, $\alpha = 0.75$, $\omega = 0.05$. NB: This violates $\mathbb{E}(A) = 1$ but amounts to a simple rescaling of the return function. Panel A: γ^{col} is denoted by the blue circle; γ^{nash} (for the two-country case) is denoted by the red square at the intersection of the best response curves (i.e. the red dashed lines); γ^{nash} (for the atomistic-countries case) is denoted by the green square further away from γ^{col} ; and $\gamma(\gamma^{col})$, and its Foreign counterpart, are denoted by the grey diamonds. Panel B: May 2023 CCyB rates against the country's share of PPP GDP of Basel Committee members (plus Norway) and a line of best fit.

therefore, corresponds to the condition under which regulators have an incentive to deviate upwards.²⁵ Now, if $\frac{dN}{d\gamma} > 0$, unless bank equity capital is supplied perfectly elastically (which would correspond to $\kappa \rightarrow 0$ in our model), it must be that $\frac{dN'}{d\gamma} < 0$.

Consistent with this prediction, there is also bank-level evidence that tightening capital requirements at home causes domestic banks to cut lending abroad (see Aiyar et al. (2014), Forbes et al. (2017)), which would also correspond to $\frac{dN'}{d\gamma} < 0$ in our model (and, as above, with an extrapolation to $\frac{dN'}{d\gamma} < 0$ and, in turn, $\frac{dN}{d\gamma} > 0$). From a broader perspective, Buch and Goldberg (2017) provide a meta analysis showing that, in general, the tightening of prudential policies (including capital requirements) spill over to generate less lending abroad.

²⁵ Interpreting bank-level estimates at the aggregate level may draw a biased picture (Berg et al., 2021). However, one would typically expect competitive pressures to result in a single bank raising proportionally less capital following an idiosyncratic requirement increase than the entire banking sector would do following a sector-wide increase.

More empirical research is required to determine the direction of aggregate equity capital flows following sector-wide changes in requirements. We also emphasise that our model predicts that the direction and intensity of the flow will vary with economic conditions. The existing empirical evidence does suggest that the direction of equity capital flows is such that higher capital requirements at Home do, on average, generate a negative externality on Foreign.

5.3. Liquidity service on deposits

In our model, capital requirements bind because deposits are effectively cheaper than equity. And deposits are cheaper because they are implicitly subsidised by deposit insurance. In practice, another reason why deposits are cheaper than equity is because they provide liquidity services (Drechsler et al., 2021, 2023). This is a source of rents for banks, and the associated deposit franchise is an important element of bank decision making. In this subsection, we show that all the main mechanisms of the model go through if capital requirements are binding because of deposit liquidity services instead of deposit insurance.

In this *liquidity service model*, households derive utility from the quantity of deposits held in domestic banks. This liquidity service is non-pecuniary, is marginally decreasing, and is independent of the financial return on deposits. Specifically, in a similar vein to [Begeau \(2020\)](#), we use the following utility function for the representative household in Home:

$$E[C] + \frac{D^{1-\eta}}{1-\eta},$$

where $\eta > 0$, C is date-1 consumption and D is date-0 deposit holdings; and similarly for that in Foreign.

These deposits are not insured and there are two segmented deposit markets: one in Home and one in Foreign. We impose that a bank raising deposits in a country lends in it. We also impose that the bank does not invest in storage (we come back to this assumption below). Hence, banks compete internationally for equity capital, and then choose in which country to operate.

Household maximisation with respect to deposit holding provides a condition which defines an inverse supply curve for (Home) deposits:

$$\mathbb{E}[R^d] = 1 - D^{-\eta},$$

where $\mathbb{E}[R^d]$ is the expected pecuniary gross return.²⁶ Note that the functional form ensures that, in equilibrium, $D > 0$ and $\mathbb{E}[R^d] < 1$, so that deposits are indeed cheaper than equity capital and the capital requirement binds. Hence, as before, we have $D^* = \frac{(1-\gamma)}{\gamma} N^*$ and $X^* = \frac{N^*}{\gamma}$. Repeating the steps of our analysis in Section 3.1 and recalling $\mathbb{E}[A]=1$, Home banks' expected return on equity is now given by

$$R(N^*, \gamma) = \frac{\alpha \left(\frac{N^*}{\gamma}\right)^{\alpha-1} - (1-\gamma) \left(1 - \left(\frac{(1-\gamma)}{\gamma} N^*\right)^{-\eta}\right)}{\gamma}. \quad (19)$$

The bank's objective is otherwise identical to Eq. (3) in our model. The analysis of Section 3.2 is unaffected by this redefinition of the return on equity. Changes in capital requirements at Home still generate a capital flow in the form described in [Lemma 2](#) and this generates a spillover on Foreign as in [Proposition 2](#).²⁷ As a result:

Proposition 5. *In the liquidity service model: (i) there also exists a unique equilibrium as defined in [Proposition 1](#) using the return on equity defined in Eq. (19) and (ii) [Theorem 1](#) also holds under [Condition 1](#).*

[Theorem 1](#) also holds in this setting.²⁸ The economic intuition is similar to that in Section 3.3, but with an added twist. As in our baseline model, the concavity of the production function means that banks face a downward sloping, convex demand curve for loans. Hence, constraining quantities by tightening the capital requirement (holding the bank's cost of funds fixed) will raise the return on equity up to a point after which returns start to decline. This follows the intuition of the fictitious monopolist bank outlined previously. However, in the liquidity service model, the representative bank also faces an upward sloping, concave supply curve for deposits. So, we can also build intuition by thinking of a fictitious *monopsonist* bank. Both the monopolist and monopsonist rents are concave in quantities. Likewise, both lending and deposits are shrinking in the capital requirement. Hence, there exists a capital requirement which maximises the combined rent and hence the return on equity. As in our baseline model, an increase in γ beyond this point causes a fall in the return on equity and hence a capital outflow.

²⁶ Note that deposits are risky but agents are risk neutral and only need to break even in expectation.

²⁷ The proofs of [Lemma 2](#) and [Proposition 2](#) depend on the partial derivatives of $R(N, \gamma)$ (and $R'(N', \gamma')$) existing and being continuous in their arguments. This is trivial to verify.

²⁸ This relies on regularity condition 1 being satisfied. If $\eta \leq 1$, then this is always true. But if deposit supply is more elastic the condition is not guaranteed to hold, so we still need to impose it. Given this condition, proving the proposition just requires following the same steps as the proofs of [Proposition 1](#) and [Theorem 1](#) with the modified expression for the return on equity.

The policy game in the liquidity service model. We now embed the liquidity service model into the policy game. For tractability, we consider the case where $\kappa \rightarrow \infty$ and A follows a binary distribution: $A \in \{0, \frac{1}{q}\}$ with $\Pr(A = \frac{1}{q}) = q$. As before, welfare is the value added of the banking system less deadweight losses. The difference is that the banking sector surplus now includes liquidity services. We have: $\pi = X^\alpha - X + \frac{(X-N)^{1-\eta}}{1-\eta} - \lambda(1-q)(X-N)$. Allowing banks to lever up generates an additional gain as liquidity service provision expands.

The problem of the collaborative regulators can be formulated as

$$X^{\text{col}} = \arg \max_X X^\alpha - X + \frac{\left(X - \frac{\omega}{2}\right)^{1-\eta}}{1-\eta} - \lambda(1-q) \left(X - \frac{\omega}{2}\right).$$

This yields the following first order condition

$$\alpha (X^{\text{col}})^{\alpha-1} - 1 = \lambda(1-q) - \left(X^{\text{col}} - \frac{\omega}{2}\right)^{-\eta}. \quad (20)$$

Compared to collaborative first order condition (14) in Section 4, the right side of Eq. (20) has an additional term reflecting that balance sheet expansion generates greater liquidity service at the margin, making lending more attractive to the regulators.

Now, under Nash competition, the Home regulator first order condition is

$$\begin{aligned} \pi_\gamma^*(\gamma, \gamma') &= \left(\alpha (X^*)^{\alpha-1} - 1 + \left(X^* - \frac{\omega}{2}\right)^{-\eta} - \lambda(1-q) \right) \frac{dX^*}{d\gamma} \\ &\quad + \left(\lambda(1-q) - \left(X^* - \frac{\omega}{2}\right)^{-\eta} \right) \frac{dN^*}{d\gamma} = 0. \end{aligned}$$

So the externality at the collaborative optimum is given by

$$\pi_\gamma^*(\gamma^{\text{col}}, \gamma^{\text{col}}) = \left(\underbrace{\lambda(1-q) - \left(X^{\text{col}} - \frac{\omega}{2}\right)^{-\eta}}_{=\alpha(X^{\text{col}})^{\alpha-1} - 1} \right) \frac{dN^*}{d\gamma}.$$

The term $\lambda(1-q) - \left(X^{\text{col}} - \frac{\omega}{2}\right)^{-\eta}$ is the marginal social value of bank capital at the collaborative optimum. From Eq. (20), it is equal to the net present (private) value of the marginal loan at the collaborative optimum. For the moment, we will proceed assuming it is positive and discuss its sign further below.

Under that assumption, the sign of the externality is once again pinned down by $\frac{dN^*}{d\gamma}$ and therefore depends on whether X^{col} is greater than, or less than, $\bar{X}$, the return maximising level of lending, where $\bar{X}$ is the level of lending that sets $R_\gamma = 0$, that is²⁹

$$\alpha^2 (\bar{X})^{\alpha-1} + (1-\eta) \left(\bar{X} - \frac{\omega}{2}\right)^{-\eta} = 1.$$

Since $\eta > 0$, it directly follows that $X^{\text{col}} > \bar{X}$ when $\lambda = 0$. As in the original model, return maximisation is associated with monopolist marginalisation. Hence, the α^2 term appears in the equation for $\bar{X}$. What changes compared to Section 4.3.1 is that the factor $\frac{1}{q}$ (which emanated from the rents associated with deposit insurance) is replaced by an additive term associated with monopsonist marginalisation. Since both monopolistic and monopsonistic marginalisation push in the same direction (lower quantities than what maximises social welfare), we have that $X^{\text{col}} > \bar{X}$. Deadweight losses still generate a force that reduces X^{col} relative to $\bar{X}$. Hence, adding a deposit franchise that yields rents for the banking sector generates similar insights to having rents from a lending franchise. A reduction in the deposit supply elasticity, η ,

²⁹ This step relies on:

$$R_\gamma(N, \gamma) = \frac{-1}{\gamma^2} \left(\alpha^2 \left(\frac{N}{\gamma}\right)^{\alpha-1} - 1 + (1-\eta) \left(\frac{1}{\gamma} - 1\right)^{-\eta} N^{-\eta} \right)$$

and $X = \frac{N}{\gamma}$, $X - \frac{\omega}{2} = \left(\frac{1}{\gamma} - 1\right) N$ in a symmetric equilibrium with fixed equity supply.

increases the likelihood that $\frac{dN^*}{d\gamma} < 0$, in a similar manner to a reduction in the loan demand elasticity, α .

Note also that X^{col} is increasing in ω and that, for $\eta > 1$, $\bar{X}$ is decreasing in ω . Therefore, the same intuition around [Proposition 4](#) still holds and so higher ω makes it more likely $\gamma^{\text{col}} < \gamma^{\text{nash}}$ and vice versa.³⁰

A difference between our main model and the liquidity service model is that, in the latter, whether $\alpha (X^{\text{col}})^{\alpha-1} - 1 > 0$, depends on parameters. If deposit supply is sufficiently inelastic, the sign can switch as the regulator sees a high marginal benefit of more deposits. Equity capital funding is then, at the margin, a welfare reducing source of funds. Such a situation may however, have limited empirical relevance. The post-crisis policy debate surrounding capital regulation has focused on the trade off between tighter regulation and lending, not tighter regulation and banks ability to provide deposits (see for example [BCBS \(2022\)](#), Chapter 8). Second, in reality, banks have access to safe assets such as reserves and government bonds that do not generate a capital charge. Here, we assume banks cannot invest in the storage technology (which carries a zero capital requirement). If we dropped this assumption, banks would raise deposits and invest in storage until any markdown on deposits had been competed away. Capital regulation would then only be relevant for lending. Reintroducing deposit insurance would take us back to our main model.

The bottom line here is that our findings do not hinge on the reason why deposits have a funding advantage. However, from a modelling perspective, the benefit of considering deposit insurance, or government guarantees in general, is that they give banks an incentive to specialise, and so their portfolio choice is determined endogenously. Absent deposit insurance, we would require the explicit, somewhat arbitrary, assumption that an individual bank will always raise deposits and lend in the same country.

6. Conclusion

Under a host-country rule for bank capital regulation, the relevant strategic interaction becomes competition for scarce bank equity capital. Depending on economic conditions, a rise in a given country's capital requirement can generate capital flows of either sign. Outflows from that country are associated with a positive externality on other countries, inflows with a negative one.

A key policy implication of our results is that this capital flow externality is likely to make CCyBs overshoot relative to what is socially optimal. Individual regulators raise requirements too much (compared to collaboration) in normal and good times and cut them too much in bad times.

An important ingredient in our analysis is that bank equity capital is supplied somewhat inelastically and equity markets are integrated between countries. This generates the inflows and outflows described above. If bank equity capital were supplied perfectly elastically, the gains from coordination would vanish. The elasticity of the global bank equity supply is an important object as it governs gains from coordination. Future research estimating this parameter could prove fruitful.

Likewise, the gains from collaboration would vanish if we endow the regulators with as many instruments as target variables (two in our case, X^* and N^*); an application of the [Tinbergen \(1952\)](#) principle. For instance, if a regulator could subsidise (without additional distortion) the allocation of bank capital to their country, it could both reduce leverage (via the capital requirement) and avoid the associated contraction in economic activity (via the subsidy). The assumption that such subsidy is, otherwise, not distortive is however rather extreme, and unlikely to be satisfied in practice. More generally, this feature of our model relates to [Korinek \(2016\)](#), which shows that violations of the Tinbergen principle can create scope for collaboration.³¹

³⁰ Even if $\eta < 1$, $\bar{X}$ is typically increasing in ω at a slower rate than X^{col} which generates the same insight.

³¹ [Korinek \(2016\)](#) identifies three conditions under which there are no gains from international policy cooperation: (i) Regulators are price takers;

CRedit authorship contribution statement

Saleem Bahaj: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Conceptualization. **Frederic Malherbe:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

Nothing to disclose.

Data availability

[The Cross-border Effects of Bank Capital Regulation: Replication Files \(Original data\)](#) (Mendeley Data)

Appendix A. The Basel III capital framework

In this appendix, we further explain why the Basel III regime can be approximated as a capital requirement that has a fixed, common component and a time-varying one set by the country where the lending takes place. Since the time varying requirement is what matters at the margin, this maps to the setting we consider in our model.

At the outset, note that the 27 jurisdictions that form the Basel Committee cover most of the world's major economic and financial centres (including the US, EU and China). These jurisdictions are compliant with Basel III regulations. For the rest of the world, Basel compliance varies. Our model should be seen as capturing competition between regulators in developed, integrated economies where capital can flow freely and so the Committee members are the best real world analogue.

General principles. Basel III prescribes a series of additive capital requirements, often also called buffers. For simplicity, we express capital requirements in terms of ratio of Tier 1 capital to risk weighted assets.³² Basel III specifies that all banks must adhere to a minimum 6% capital requirement at all times, plus an additional 2.5% capital conservation buffer that the bank can use temporarily to absorb a loss but will face dividend restrictions as a result. The minimum requirement and conservation buffer are homogeneous across jurisdictions and lending location and are not time varying. We view the parameter $\underline{\gamma}$ in Eq. (18) as jointly capturing these minima.

The CCyB. As described in Section 5.1, the static requirements are supplemented by the countercyclical capital buffer (CCyB). The CCyB has a macroprudential purpose. It is time varying by design. We elaborate more on its features here.

The CCyB is set at the discretion of regulators in a particular jurisdiction. In particular, it is set on lending (or other bank exposures) to agents within the regulator's jurisdiction. The regulator can always impose the requirement on banks within its jurisdiction (domestic

(ii) the Tinbergen Principle is satisfied (for external instruments); and (iii) International Markets are frictionless. The third condition is satisfied in our model. The second is not, as we have discussed. Neither is the first, but the existence of gains from collaboration does not hinge on this violation. In fact, gains from cooperation increase when regulators are price takers (see Section 4.4).

³² Tier 1 capital includes some contingent capital instruments beyond common equity that enable the bank to absorb losses on a going concern basis. The minimum capital requirement for common equity is 1.5 percentage points lower. The Basel III framework also contains an additional 2 percentage point requirement for Tier 2 capital, which can be thought of as gone-concern loss absorbing liabilities such as subordinated debt. Our model makes no distinction between common equity and other forms of contingent liabilities so we abstract from different definitions of capital.

banks and subsidiaries of foreign banks). For banks outside the regulator's jurisdiction that may lend within it, the framework is based on obligatory reciprocity: foreign regulators must impose the equivalent CCyB on the banks they regulate. For members of the Basel committee the obligation to reciprocate is encoded in the law governing capital regulation.

In relation to our analysis, reciprocity is a crucial dimension of the regulatory framework. In short, even though, *de jure* the rule is home-country based, *de facto* it is a host-country rule. This is why we argue that the Basel III regime can be approximated as a capital requirement that has a fixed, common component and a time-varying one set by the country where the lending takes place.

This is what leads to the capital requirement we described in Eq. (18). We repeat it here for convenience

$$n_{it} \geq (\gamma + \text{CCyB}_i)x_{it} + (\gamma + \text{CCyB}'_i)x'_{it}.$$

Let us describe how this capital requirement plays out in practice with an example. Assume Home is the UK and Foreign is Germany. Imagine a bank, i , under the UK's jurisdiction has equity n_{it} and lends x_{it} in the UK and x'_{it} in Germany. Imagine that UK and German regulators adhere to a static common capital requirement, γ , representing the minimum requirement and conservation buffer. The UK regulator sets CCyB_i on lending in the UK. This is the marginal requirement where regulators have discretion. The German regulator sets CCyB'_i on German lending. This means per unit of UK lending the UK regulator ensures bank i must have at least $(\gamma + \text{CCyB}_i)$ of equity capital. Reciprocity means that when the German regulator announces CCyB'_i , the UK regulator as part of its supervision of bank i also ensures it has at least $(\gamma + \text{CCyB}'_i)$ equity capital per unit of German lending.

This regulatory arrangement maps into that of our model if one sets $\gamma = \gamma + \text{CCyB}_i$, except that we are abstracting from the bounds: in reality the CCyB cannot be negative and mandatory reciprocity only applies up to 2.5%; thereafter it is voluntary. We explain in Section 5.2 how this affects the implications of our model.

Bank-specific requirements. Regulators also have the option to set bank specific capital requirements (that apply to all the loans the bank makes, irrespective of the location of the borrower). To fix ideas, one can incorporate these into Eq. (18) in the following manner

$$n_{it} \geq (\gamma + \text{CCyB}_i)x_{it} + (\gamma + \text{CCyB}'_i)x'_{it} + \gamma_i(x_i + x'_i),$$

with $\gamma_i \geq 0$. These requirements may be used because the bank is viewed as systemic or has raised a regulatory flag (e.g. failing a stress test). The banks in our model are ex-ante identical and where they lend is the only way in which they differ in equilibrium. Hence there is no role for a bank specific capital requirement.

We believe there are two reasons why that abstracting from bank specific requirements in the context of our analysis is a valid approach. First, from the perspective of international coordination, the most relevant bank specific capital requirements is the buffer imposed on Globally Systemically Important Banks (G-SIBs). This buffer is, technically, implemented by the national regulator but the calculation for the size of the buffer is formulaic and agreed by the Basel Committee eliminating regulators' discretion. Since G-SIB buffers are coordinated there is a good case for thinking of it as part of the common standard, meaning that we return to a situation approaching Eq. (18). The loss of tractability from considering coordinated but bank-specific requirements is unlikely to be worth the increase in generality. Second, γ_i is bounded below by 0. If other bank-specific capital requirements, with regulatory discretion, suffer from a race-to-the-bottom (as in Dell'Ariccia and Marquez (2006)), one could expect these to be set to 0 anyway, again returning to Eq. (18).³³ More generally, we argue

³³ In practice, there is a degree of coordination on most bank specific capital requirements. For instance, the most systemic banks within the EU have their domestic systemic institution buffers set by the union-wide regulator rather than national regulators. If all bank specific requirements were coordinated this would bring us back to the G-SIB case.

that there is no reason to believe that the presence of bank specific requirements eliminates the potential for strategic behaviour regarding the setting of the CCyB.

Sector-specific requirements. Regulators have scope to set additional capital requirements on specific types of lending for macroprudential purposes. These are known as sectoral capital requirements. So for example, a regulator could raise the capital requirement on a particular type of mortgage lending within its jurisdiction. There is no mandatory reciprocation of sectoral requirements. Interestingly, some jurisdictions or groups of jurisdictions have set up voluntary reciprocation regimes. The ESRB is a particular example of this: see [here](#) for a description of their regime. Voluntary reciprocation is not codified in the same way as the mandatory reciprocation of the CCyB. Nonetheless, moral suasion, reputational costs and repeated game aspects appear to sustain equilibria where regulators do agree to reciprocate sectoral requirements in practice. In our model, each country has a single sector, so there would be no role for sector-specific requirements.

Implementation: supervisory capture. Implementing a capital requirement also requires supervisors to calculate n_{it} (raising issues regarding the definition of regulatory capital or the rate of bank loss recognition) and x_{it} (involving the calculation of risk weights or the treatment of off-balance sheet exposures). These calculations are complex and there is potential for interpretation. One can imagine scenarios in which supervisors are captured by the domestic banking sector lobby. In such a case, national supervisors may choose to be lenient (e.g., by letting banks in its jurisdictions have artificially low risk weights), which could lead to a scenario reminiscent of the race to the bottom in Dell'Ariccia and Marquez (2006). Studying the interaction between supervisory forbearance and our mechanisms is potentially interesting, but is beyond the scope of this paper.

Appendix B. Additional theoretical results

B.1. Higher capital requirements make banks strictly worse off

Here we relax the perfect competition assumption and extend the model in Section 3 to allow banks to have market power in the loan market. In particular, we assume that there are ν identical banks specialising in lending each country and compete a la Cournot to supply loans to penniless firms.³⁴ Banks still act as price takers in the markets for deposits and equity capital. We focus on banks specialising in Home lending and abstract from the Foreign country (function dependencies on γ' are omitted). Our goal is to show that a rise in capital requirements strictly decreases the pure profits of banks in equilibrium.

To briefly summarise the environment: as before, banks fund their lending by raising a mix of deposits and equity. Deposits are insured and are raised elastically from households that have an opportunity cost of funds of one. For the purposes of this extension we assume each bank has a vanishingly small endowment of inside equity. The bank can raise new outside equity from the global capital market. As in the main text, outside equity must return in expectation, $1 + z = 1 + \frac{z}{2} M^2$ per unit raised, where M is the total amount of new outside equity raised both in Home and Foreign. Banks take z as given. The regulator sets the minimum share of equity financing (inside and outside) in the form of the capital requirement, γ . This requirement always binds in equilibrium and is taken as a parameter in the spirit of Section 3.

As in the baseline model, firms borrow from banks to invest in physical capital that is used to produce consumption goods using an

³⁴ From Schliephake and Kirstein (2013) this is equivalent to banks first raising equity to generate lending capacity and then, holding fixed the liability side of the balance sheet, competing a la Bertrand (see also Kreps and Scheinkman (1983)).

aggregate Cobb–Douglas technology with curvature parameter $0 < \alpha < 1$. For simplicity, we assume that aggregate TFP is binary: it is zero with probability $(1 - q)$ and equal to $\frac{1}{q}$ with probability q .

Consider bank i that makes loans $x_i = \frac{n_i}{\gamma}$, where n_i is the value of the firm's equity. Since firms are penniless all the returns on physical capital are captured by the bank. Hence, bank expected revenues are given by $\alpha \left(x_i + \sum_{j \neq i} x_j \right)^{\alpha-1} x_i$. Imposing binding capital requirements, the bank's problem can be framed as a choice of how much equity to raise in order to maximise pure profits or, equivalently, the cash flows that accrue to inside equity:

$$\max_{n_i} \frac{n_i}{\gamma} \left(\alpha \left(\frac{n_i + \sum_{j \neq i} n_j}{\gamma} \right)^{\alpha-1} - q(1 - \gamma) \right) - n_i(1 + z).$$

The first order condition is given by

$$\frac{1}{\gamma} \left(\alpha \left(\frac{n_i + \sum_{j \neq i} n_j}{\gamma} \right)^{\alpha-1} - q(1 - \gamma) \right) + \frac{n_i}{\gamma} \left(\alpha \frac{1}{\gamma} (\alpha - 1) \left(\frac{n_i + \sum_{j \neq i} n_j}{\gamma} \right)^{\alpha-2} \right) = (1 + z).$$

In equilibrium banks are symmetric so $n_i = n_j = n^*(\gamma)$. Hence, we obtain the standard result that banks demand a return on loans that is a fixed markup over a term that can be interpreted as a marginal cost:

$$\underbrace{\alpha \left(\frac{vn^*(\gamma)}{\gamma} \right)^{\alpha-1}}_{\text{return on loans}} = \underbrace{\left(1 + \frac{1 - \alpha}{\alpha + (v - 1)} \right)}_{\text{markup} \equiv \mu} \underbrace{\left(\gamma(1 + z^*(\gamma)) + (1 - \gamma)q \right)}_{\text{marginal cost} \equiv \text{MC}^*(\gamma)} \quad (\text{B.1})$$

In Eq. (B.1), $1 + z^*(\gamma)$ is the equilibrium cost of outside equity. At the same time, deposits are only repaid by the bank with probability q at a gross rate of unity. Given γ is the portion of equity (versus deposit) finance, this leads to the intuitive expression for marginal cost of funds: $\text{MC}^*(\gamma) = \gamma(1 + z^*(\gamma)) + (1 - \gamma)q$.

Given this markup, pure profits for an individual bank are

$$\pi^*(\gamma) = \mu \text{MC}^*(\gamma) x^*(\gamma),$$

where $x^*(\gamma)$ is the equilibrium level of lending. We can now show that π^* decreases with γ . We have that

$$\frac{d\pi^*}{d\gamma} = \mu \left(\frac{dx^*}{d\gamma} \text{MC}^*(\gamma) + \frac{d\text{MC}^*}{d\gamma} x^*(\gamma) \right).$$

Noting $x^*(\gamma) = \frac{n^*(\gamma)}{\gamma}$ and applying the implicit function theorem to Eq. (B.1) we obtain

$$\frac{dx^*}{d\gamma} = \frac{\frac{d\text{MC}^*}{d\gamma} x^*(\gamma)}{(\alpha - 1)\text{MC}^*(\gamma)}.$$

Hence

$$\frac{d\pi^*}{d\gamma} = \mu \frac{d\text{MC}^*}{d\gamma} x^*(\gamma) \underbrace{\left(\frac{\alpha}{\alpha - 1} \right)}_{< 0}.$$

So, it is sufficient to show that $\frac{d\text{MC}^*}{d\gamma}$ is always positive. Now,

$$\frac{d\text{MC}^*}{d\gamma} = 1 + z^*(\gamma) - q + \kappa \gamma M^*(\gamma) \frac{dM^*}{d\gamma}.$$

Using Proposition 2 and noting that $N^*(\gamma) = v\gamma x^*$, we have

$$\frac{dM^*}{d\gamma} = \frac{dN^*}{d\gamma} + \frac{dN^{f*}}{d\gamma} = v(1 - \text{SP}^*) \left(x^*(\gamma) + \gamma \frac{dx^*}{d\gamma} \right).$$

Note that to appeal to Proposition 2, we just require that investors in outside equity equate the return on outside equity to $1 + z$ in both countries. We do not need to specify the competitive environment in Foreign.

As a result, we have that

$$\frac{d\text{MC}^*}{d\gamma} = 1 + z^*(\gamma) - q + \kappa \gamma v M^*(1 - \text{SP}^*) \left(x^* + \gamma \frac{dx^*}{d\gamma} \right),$$

or

$$\frac{d\text{MC}^*}{d\gamma} = \left(1 + \frac{\kappa \gamma^2 v M^*(1 - \text{SP}^*)}{(1 - \alpha)\text{MC}^*(\gamma)} \right)^{-1} (1 + z^*(\gamma) - q + \kappa \gamma v M^*(1 - \text{SP}^*) x^*).$$

Inspecting terms it is straightforward to verify that MC^* is always increasing in γ .

To give the intuition for this result, note that an increase in γ affects MC^* through two channels. First, it causes a switch in the composition of liabilities from deposits to capital. As the latter is a more expensive form of finance from the Bank's point of view this raises marginal cost. Second, the change in the capital requirement generates a change in the demand for equity capital. When the increase in requirements causes banks to raise more equity, this simply causes a further increase in marginal cost. When the increase in requirements causes banks to raise less equity, this mitigates the increase in MC^* associated with the composition effect, but cannot overturn it, otherwise we would get a contradiction: there would be higher capital requirements, higher lending, yet less equity in equilibrium (this also follows from the proof of Lemma C.3 below).

In turn, this implies Bank pure profits are strictly decreasing in the capital requirement. The markup means that while higher costs can be offset by higher revenues, the offset can never be complete. Moreover, the decline in profits is increasing in the markup hence a less competitive banking sector would be more averse to an increase in capital requirements.

B.2. Generalising the welfare analysis

In the main text, we considered a welfare function $\pi^*(\gamma, \gamma')$ with a specific functional form. Here we first generalise to the following class of objective functions

$$\pi^*(\gamma, \gamma') = V(N^*(\gamma, \gamma'), \gamma). \quad (\text{B.2})$$

This specification says that Home capital requirements can affect Home welfare in an arbitrary way. However, Foreign capital requirements matter only to the extent they affect N , which, along with γ , pins down the volume of credit at Home.

The capital flow externality on Foreign following a Home capital requirement change can be expressed as:

$$\pi_{\gamma'}^*(\gamma', \gamma) = \frac{dN^{f*}}{d\gamma} V'_{N'}(N^{f*}, \gamma'). \quad (\text{B.3})$$

Eq. (B.3) can be interpreted as the equilibrium marginal social value of capital in Foreign, $V'_{N'}(N^{f*}, \gamma') > 0$, multiplied by the capital flow from Foreign following a change in the Home requirement, $\frac{dN^{f*}}{d\gamma}$. So, the main logic of our analysis goes through if in equilibrium equity capital is valuable at the margin (specifically, $V_N(N^{\text{col}}, \gamma^{\text{col}}) > 0$ and $V_N(N^{\text{nash}}, \gamma^{\text{nash}}) > 0$). This will be the case for a broad range of objective functions:

Proposition 6. For any objective $\pi^*(\gamma, \gamma') = V(N^*(\gamma, \gamma'), \gamma)$, if $V_N(N^{\text{nash}}, \gamma^{\text{nash}}) > 0$, then Proposition 3 holds.

A sufficient condition for $V_N(N^{\text{nash}}, \gamma^{\text{nash}}) > 0$ is that, given a level of lending X , reducing leverage improves social welfare. This is the case in our policy game since $S_N(X, N) > 0$ as long as banks default with positive probability. We can generalise this as follows:

Proposition 7. Define $\tilde{V}(X, N) \equiv V(N, \gamma = N/X)$. Then $\tilde{V}_N(X, N) > 0, \forall N < X$ is a sufficient condition for Proposition 3 to hold under general objective (B.2).

The intuition for Proposition 7 is that if the regulator always benefits from a reduction in leverage holding lending fixed, then lending must be socially valuable at the regulator's optimum (otherwise the regulator would reduce leverage). Therefore, an increase in N holding γ fixed is welfare enhancing in equilibrium as it would lead to an expansion of lending. And so we have $V'_{N'}(N^*, \gamma') > 0$. The net result is that a capital inflow is always socially valuable at the relevant optima.

B.2.1. Generalising further

We now put our results in a broader perspective by considering how regulatory incentives change in our model when the objective function allows for changes in γ to affect π' in other ways than through its effect on N^* . To capture this, we rewrite the objective as $V(N^*, \gamma, \gamma')$ where the final argument captures all mechanisms via which the capital requirements in foreign directly affect welfare in Home (that is, beyond its impact on N^*).

The generalised externality then takes the form:

$$\pi_\gamma^*(\gamma', \gamma) = \frac{dN^*}{d\gamma} V'_{N'}(N^*, \gamma', \gamma) + V'_\gamma(N^*, \gamma', \gamma). \quad (\text{B.4})$$

Three points are in order. First, generalising the objective function does not affect the existence of the capital flow externality: the first term in the right hand side of the above equation is essentially identical to that in our model. Furthermore, the sign of $\frac{dN^*}{d\gamma}$ still hinges here on whether $\gamma \gtrless \hat{\gamma}(\gamma')$: if $R_\gamma(N^*, \gamma)$ is positive, an increase in γ siphons off capital from Foreign. As long as $V'_{N'}(N^*, \gamma', \gamma)$ is positive this constitutes a negative externality.

Second, that an increase in bank capital, holding lending fixed, increases welfare (i.e., the counterpart to our condition $\tilde{V}_N(X, N) > 0$ in Proposition 7) would no longer be a sufficient condition for $V'_N > 0$.

Third, there also is a "direct" capital requirement externality $V'_\gamma(N^*, \gamma', \gamma)$. For instance this could capture that a decrease in Home bank's probability of default has a positive impact on welfare in Foreign. Obviously, in the presence of this additional term, the sign of the total externality no longer hinges on that of $\frac{dN^*}{d\gamma}$. The additional externality may either reinforce, mitigate, or even offset the bank equity capital flow externality.

One case where the objective is of this form is when regulators treat equity capital as socially costly. We sketch this case next.

B.2.2. The externality when regulators treat equity capital as socially costly

Adding the cost of equity capital to our welfare function requires making an assumption on how the initial wealth ω is allocated across countries. For simplicity, we assume it is split equally. Adapting Eq. (11) accordingly yields

$$\pi^*(\gamma, \gamma') \equiv (X^*)^\alpha - X^* - \lambda S(X^*, N^*) - z \underbrace{(N^*, N^*)}_{\text{total cost of equity}} \left(N^* - \frac{\omega}{2} \right),$$

where the term $z(N^*, N^*) \left(N^* - \frac{\omega}{2} \right)$ reflects the excess cost of the equity used by the banks specialising in Home lending, less the excess return Home investors receive on their initial wealth. Recalling, $z(N^*) = \frac{\kappa}{2} (\max \{ N^*(\gamma, \gamma') + N^*(\gamma', \gamma) - \omega, 0 \})^2$, so that $z^*_{N'} = z^*_{N'} = \kappa M^*$, and $\frac{dN^*}{d\gamma} = \text{SP}^* \frac{dN^*}{d\gamma}$, the collaborative FOC becomes

$$\underbrace{\text{NPV}^* \frac{dX^*}{d\gamma} - \left(\kappa M^* (1 - \text{SP}^*) \left(N^* - \frac{\omega}{2} \right) + z^* + \lambda S^*_{N'} \right) \frac{dN^*}{d\gamma}}_{=\pi_\gamma^*(\gamma, \gamma')} + \text{NPV}^* \frac{dX^*}{d\gamma} - \left(\kappa M^* \left(1 - \frac{1}{\text{SP}^*} \right) \left(N^* - \frac{\omega}{2} \right) + z^* + \lambda S^*_{N'} \right) \frac{dN^*}{d\gamma} = 0, \\ \underbrace{\hspace{10em}}_{=\pi_\gamma^*(\gamma', \gamma)}$$

where for compactness we retain the notation $\text{NPV}^* = \alpha (X^*)^{\alpha-1} - 1 - \lambda S^*_X$ (note, the economic interpretation of this object is not the same as in Section 4 since the true NPV computation would require discounting at a different rate). Since capital requirements bind we have

$$\frac{dX^*}{d\gamma} = \frac{1}{\gamma} \frac{dN^*}{d\gamma} - \frac{X^*}{\gamma} \& \frac{dX^*}{d\gamma} = \frac{1}{\gamma'} \frac{dN^*}{d\gamma}.$$

Hence, the externality is given by

$$\pi_\gamma^*(\gamma', \gamma) = \left(\frac{\text{NPV}^*}{\gamma'} - \lambda S^*_{N'} \right) \frac{dN^*}{d\gamma} - \frac{dN^*}{d\gamma} z^* \\ - \kappa M^* \times \underbrace{\left(\frac{\text{SP}^* - 1}{\text{SP}^*} \right) \frac{dN^*}{d\gamma}}_{\equiv \frac{dN^*}{d\gamma} + \frac{dN^*}{d\gamma}} \times \left(N^* - \frac{\omega}{2} \right). \quad (\text{B.5})$$

Now we see that externality has three terms each containing a factor $\frac{dN^*}{d\gamma}$. The first term corresponds exactly to the externality of our main analysis, that is with an objective given by Eq. (11). That z^* is internalised does not alter that $\left(\frac{\text{NPV}^*}{\gamma'} - \lambda S^*_{N'} \right)$ is positive at the collaborative optimum.

The second and third term captures all additional effects running through the cost of equity. The second term reflects that a marginal unit of new equity has an excess cost z^* ; so if $\frac{dN^*}{d\gamma}$ is positive there is a direct negative consequence on Foreign. The third term reflects that, in equilibrium, investing a marginal unit in Foreign in response to a change in γ would require global demand for new equity to fall by a factor $\frac{1 - \text{SP}^*}{\text{SP}^*}$; this lowers z^* , benefiting Foreign. On balance then, these additional forces are ambiguous in sign.

It turns out the extent of spillovers ends up governing this sign. To see this, recall that $\frac{\kappa}{2} M^{*2} = z^*$ and that $N^* - \frac{\omega}{2} = \frac{M^*}{2}$ at any symmetric point. So, the externality at the collaborative optimum boils down to

$$\pi_\gamma^*(\gamma^{\text{col}}, \gamma^{\text{col}}) = \left(\frac{\text{NPV}^{\text{col}}}{\gamma'} - \lambda S^{\text{col}}_{N'} \right) \frac{dN^*}{d\gamma} \\ + \frac{\kappa}{2} (M^{\text{col}}) \frac{(1 - 2\text{SP}^{\text{col}})}{\text{SP}^{\text{col}}} \frac{dN^*}{d\gamma}.$$

The additional terms relating to the cost of equity reinforce the capital flow externality whenever $\text{SP}^{\text{col}} < \frac{1}{2}$, otherwise they are a countervailing force, which can potentially dominate. We do know, however, that when capital is relatively abundant, i.e. ω is large, spillovers are smaller as less equity is being raised and the equilibrium is on a more elastic portion of the global supply curve. We also know that the larger ω the more likely $\frac{dN^*}{d\gamma} < 0$ (see Proposition 4). So internalising the cost of equity would reinforce the race to the top when capital is abundant. (In contrast, there may be a moderation in the race to the bottom when there is capital scarcity).

While analysing the case where the cost of equity is internalised yields economic forces and expressions that are interpretable, it is challenging to map the mechanisms into real world counterparts. For example, the third term in Eq. (B.5) captures that the Home regulator internalises the cost of equity and acts like a monopsonist in the global equity market to lower this cost. However, Foreign benefits from lower costs too and so the competitive Home regulator ends up demanding too much capital in the Nash equilibrium. This generates a force towards a race to the top. But the manipulation of global equity markets is not in line with regulatory behaviour in practice. To fully understand the implications of regulators internalising the cost of equity, microfoundations on the source of the cost would be needed. These are beyond the scope of this paper.

Overall, the key point that emerges from this analysis is that the mechanisms described in Section 4 under objective (11) are still present even when the regulator internalises the cost of equity. In this case, what we see are additional forces that may mitigate or reinforce the externality arising from capital flows identified in the main text.

Appendix C. Proofs

Remark. When proofs are identical for Home and Foreign, we only provide the former.

Lemma 1. (i) Banks default with strictly positive probability in equilibrium; (ii) capital requirements are binding; and (iii) each individual bank perfectly specialises as either a lender in Home or Foreign.

Proof. Consider a bank i with equity capital n_i and for which x_i and x'_i denotes lending in Home and Foreign respectively. The bank defaults if

$$\alpha X^{\alpha-1} x_i + \alpha A' (X')^{\alpha-1} x'_i < x'_i + x_i - n_i,$$

This means we can define two functions that both represent the default boundary: points in the state space where the Bank just has sufficient resources to make depositors whole.³⁵

$$a_i^0(A') = \frac{x'_i + x_i - n_i - \alpha A' (X')^{\alpha-1} x'_i}{\alpha X^{\alpha-1} x_i},$$

$$a_i^0(A) = \frac{x'_i + x_i - n_i - \alpha X^{\alpha-1} x_i}{\alpha (X')^{\alpha-1} x'_i}.$$

Given this, we can write the bank's problem as the following Lagrangian (ignoring non-negativity constraints, and noting that $\mathbb{E}[A] = \mathbb{E}[A'] = 1$)

$$\begin{aligned} & \max_{n_i, x_i, x'_i} \left(\alpha X^{\alpha-1} x_i + \alpha (X')^{\alpha-1} x'_i - x'_i - x_i + n_i \right) + \\ & \int_0^{a_i^0(0)} \int_0^{a_i^0(A)} \left(x'_i + x_i - n_i - \alpha X^{\alpha-1} x_i - \alpha A' (X')^{\alpha-1} x'_i \right) h(A, A') dA' dA + \\ & \psi (n_i - \gamma x_i - \gamma' x'_i) - n_i z(N, N'), \end{aligned}$$

where $h(A, A')$ is the joint density of A and A' . The first row corresponds to the economic surplus generated by the bank's lending. The second row is strictly positive when the bank defaults with strictly positive probability (zero otherwise) and captures the implicit subsidy to the bank arising from the government's guarantee. The third row captures constraints arising from the capital requirements, with corresponding multiplier ψ , and the excess cost to the bank of the capital raised.

We first prove by contradiction that capital requirements must be binding in equilibrium. Imagine they are not and the multiplier is nil. Then the bank can reduce n_i holding x_i and x'_i fixed. Since x_i and x'_i are fixed, the surplus from lending (the first line) is unaffected. However, the implicit subsidy can be increased by decreasing n_i . When the subsidy is positive a marginal increase in n_i both reduces the integrand and shifts the default boundary inwards. If the subsidy is initially nil, a sufficiently large decrease in n_i will render it positive. So, the deviation is profitable. Hence, capital requirements must bind in equilibrium, (ii).

We do not consider $\gamma, \gamma' = 1$, coupled with binding capital requirements this implies in equilibrium $n_i < x_i + x'_i$. This means $a_i^0(0)$ and $a_i^0(A)$ are strictly positive in equilibrium. Since A and A' can both be nil, banks default with strictly positive probability in equilibrium, (i).

We also prove part (iii) by contradiction. Consider a hypothetical equilibrium where banks do not perfectly specialise; that is, there is an interior solution at the bank level for both x_i and x'_i . Define the following function

$$rc(A, \gamma) = \alpha X^{\alpha-1} - (1 - \gamma).$$

For given A and γ , the term rc is the residual cash flows available to the bank's shareholders from lending in Home. It corresponds to

bank revenue, net of repayments to depositors, per unit of lending in home. We can write the return, for this bank, of a unit of equity being allocated to a loan in Home as:

$$\frac{rc(1, \gamma)}{\gamma} - \int_0^{a_i^0(0)} \int_0^{a_i^0(A)} \left(\frac{rc(A, \gamma)}{\gamma} \right) h(A, A') dA' dA;$$

and similarly for a loan in Foreign.

If both x_i and x'_i are strictly positive in equilibrium, the bank must be indifferent at the margin. Then,

$$\begin{aligned} & \frac{rc(1, \gamma)}{\gamma} - \int_0^{a_i^0(0)} \int_0^{a_i^0(A)} \left(\frac{rc(A, \gamma)}{\gamma} \right) h(A, A') dA' dA = \\ & \frac{rc'(1, \gamma)}{\gamma} - \int_0^{a_i^0(0)} \int_0^{a_i^0(A)} \left(\frac{rc'(A, \gamma')}{\gamma'} \right) h(A, A') dA' dA. \end{aligned}$$

We now show that an individual bank can profitably deviate by lending entirely in the Home country. To see this first note that the after-deviation revenues of such a bank is:

$$n_i \left(\frac{rc(1, \gamma)}{\gamma} - \int_0^{A^0} \left(\frac{rc(A, \gamma)}{\gamma} \right) g(A) dA \right),$$

where, as in the text, A^0 is the default boundary for a bank that specialises in lending at Home. Making use of the fact that returns are equalised across countries allows us to compute the benefit from deviating as

$$n_i \left(\int_0^{a_i^0(0)} \int_0^{a_i^0(A)} \left(\frac{rc(A, \gamma)}{\gamma} \right) h(A, A') dA' dA - \int_0^{A^0} \left(\frac{rc(A, \gamma)}{\gamma} \right) g(A) dA \right).$$

By definition, $rc(A, \gamma) < 0$ for all $A \in [0, A^0]$. Hence, this benefit is strictly positive so long as there exists $A \in [0, a_i^0(0)]$ such that $rc(A, \gamma) > 0$. This is guaranteed by $h(A, A')$ having full support over $[0, A^H] \times [0, A'^H]$. Since a profitable deviation exists, an individual bank will never choose for both x_i and x'_i to be interior at the same time. Hence, the only possible equilibrium is one where individual banks perfectly specialise in either country and aggregate returns are equated by the aggregate amount of equity capital allocated to each country. $\square$

For further reference, we prove the following two Lemmas.

Lemma C.1. For all $\gamma, \gamma' \in (0, 1)$, and all $N, N' > 0$, the partial derivatives $R_N(N, \gamma)$ and $R'_{N'}(N', \gamma')$ exist and are continuous in their arguments.

Proof. First note that from the smoothness of $g(A)$ we have that $R(N, \gamma)$ is continuous in N . The derivative reads:

$$R_N(N, \gamma) = \begin{cases} \frac{1}{\gamma^2} \int_{A^0(N, \gamma)}^{A^H} \left(\alpha(\alpha - 1) A \left(\frac{N}{\gamma} \right)^{\alpha-2} \right) g(A) dA & A^0(N, \gamma) \leq A^H \\ 0 & A^H \leq A^0(N, \gamma). \end{cases}$$

Continuity directly follows as the integral is continuous, and the two possible values coincide at $A^0(N, \gamma) = A^H$. (Note that $\gamma > 0$ ensures $A^0(N, \gamma) > 0$ so we do not need to consider the lower bound for A). $\square$

Lemma C.2. For all $\gamma, \gamma' \in (0, 1)$, and all $N, N' > 0$, the partial derivatives $R_\gamma(N, \gamma)$ and $R'_{\gamma'}(N', \gamma')$ exist and are continuous in γ and γ' , respectively.

Proof. First note that from the smoothness of $g(A)$ we have that $R(N, \gamma)$ is continuous in γ . The derivative reads:

$$R_\gamma(N, \gamma) = \begin{cases} \frac{-1}{\gamma^2} \int_{A^0(N, \gamma)}^{A^H} \left(\alpha^2 A \left(\frac{N}{\gamma} \right)^{\alpha-1} - 1 \right) g(A) dA & A^0(N, \gamma) \leq A^H \\ 0 & A^H \leq A^0(N, \gamma). \end{cases}$$

Continuity directly follows as the integral is continuous, and the two possible values coincide at $A^0(N, \gamma) = A^H$. $\square$

Proposition 1. For all $\gamma, \gamma' \in (0, 1)$, there exists a unique pair $\{N^*, N'^*\}$ such that returns on equity in both countries are equal to the investors'

³⁵ Note that if A' is such that $a^0(A') < 0$, the bank simply cannot default for this value of A' .

required return. This pair is implicitly defined by:

$$R(N^*, \gamma) = R'(N'^*, \gamma') = (1 + z(N^*, N'^*)).$$

Proof. First, let us establish that the double equality necessarily holds in equilibrium. If $R(N^*, \gamma) > (1 + z(N^*, N'^*))$, the representative bank will scale up infinitely but this cannot be the case since $z \geq 0$, $\lim_{N \rightarrow \infty} R(N, \gamma) = 0$ and, in equilibrium, $n = N$. Likewise, If $R(N^*, \gamma) < (1 + z(N^*, N'^*))$ banks will choose $n = 0$, which is inconsistent with $n = N$ and $\lim_{N \rightarrow 0} R(N^*, \gamma) = \infty$. Hence, in equilibrium we must have that $R(N^*, \gamma) = 1 + z(N^*, N'^*)$ and, with the same logic, $R'(N'^*, \gamma') = 1 + z(N^*, N'^*)$.

It is straightforward to show that, for a given z , there is a single, interior N that solves $R(N, \gamma) = 1 + z$. But z depends on the global demand for bank equity capital: $z = \frac{\kappa}{2} (\max\{N + N' - \omega, 0\})^2$. So, the question is whether there is a unique pair (N, N') that solves the system:

$$\begin{cases} R(N, \gamma) = 1 + z(N, N') \\ R'(N', \gamma') = 1 + z(N, N') \end{cases} \quad (C.1)$$

By construction, $z \geq 0$. So, if the left-hand-side functions in system (C.1) are invertible for N and N' for values of $R, R' \geq 1$ (which we prove later), we can write the bank capital demand functions for Home and Foreign implicitly (and for a given pair $\{\gamma, \gamma'\}$) as:

$$\begin{cases} N(z) = R^{-1}(1 + z) \\ N'(z) = R'^{-1}(1 + z) \end{cases} \quad (C.2)$$

And we can aggregate them to write an implicit global demand function for bank capital: $N^D(z) = R^{-1}(1 + z) + R'^{-1}(1 + z)$.

From $z = \frac{\kappa}{2} (\max\{N + N' - \omega, 0\})^2$, we have an explicit global supply function:

$$N^S(z) \equiv \begin{cases} \frac{\sqrt{2z}}{\kappa} + \omega & , z \geq 0 \\ \omega & , z = 0 \end{cases}$$

If $N^D(0) \leq \omega$, there is excess supply,³⁶ and we simply have $z^* = 0$ and N^* and N'^* pinned down by their respective demand functions (C.2).

To complete the proof, we first show that (i) $R(N, \gamma)$ and $R'(N', \gamma')$ are respectively invertible for N and N' for values of $R, R' \geq 1$; and that (ii) $N^D(z)$ and $N^S(z)$ always cross exactly once when $N^D(0) > \omega$, with $z^* > 0$. Then, unique corresponding values of N and N'^* arise from Eqs. (C.1).

(i) From Lemma C.1, we have

$$R_N(N, \gamma) = \begin{cases} \frac{1}{\gamma^2} \int_{A^0(N, \gamma)}^{A^H} \left(\alpha(\alpha - 1) A \left(\frac{N}{\gamma} \right)^{\alpha-2} \right) g(A) dA & 0 \leq A^0(N, \gamma) < A^H \\ 0 & A^H \leq A^0(N, \gamma) \end{cases}$$

Since $R_N < 0$ when $A^0(N, \gamma) < A^H$, $R(N, \gamma)$ is invertible for N over the corresponding range of values such that $A^0(N, \gamma) < A^H$. The function is therefore invertible for the relevant range of values where $R \geq 1$. (Note that $R < 1$ cannot be an equilibrium as shareholders would always prefer to invest their wealth in the storage technology.)

(ii) $N^S(z)$ is strictly increasing starting from the point $N^S(0) = \omega$. Given (i), we know that R^{-1} and R'^{-1} are decreasing in z . Hence, $N^D(z)$ is decreasing too. If $N^D(0) > \omega$, this ensures single crossing at a point $z^* > 0$. $\square$

For further reference, we can now state:

Corollary C.1. For all $\gamma, \gamma' \in (0, 1)$ and associated $\{N^*(\gamma, \gamma'), N'^*(\gamma', \gamma)\}$, we have $R_N(N^*, \gamma) < 0$ and $R'_{N'}(N'^*, \gamma') < 0$.

³⁶ Strictly speaking, it is investors' wealth which is in excess of the demand for bank equity. Since they can use a storage technology with unit gross return, their supply is actually perfectly elastic in this region.

Lemma 2. For all $\gamma, \gamma' \in (0, 1)$,

$$\frac{dN^*}{d\gamma} = R_\gamma(N^*, \gamma) \xi^*(\gamma, \gamma'), \quad (C.3)$$

$$\text{where } \xi^*(\gamma, \gamma') \equiv \left(\frac{\kappa M^* - R'_{N'}(N'^*, \gamma')}{R_N(N^*, \gamma) R'_{N'}(N'^*, \gamma') - \kappa M^* (R'_{N'}(N'^*, \gamma') + R_N(N^*, \gamma))} \right) \Big|_{N^*(\gamma, \gamma'), N'^*(\gamma', \gamma)} > 0.$$

Therefore, it is the case that

$$\frac{dN^*}{d\gamma} \geq 0 \Leftrightarrow R_\gamma(N^*, \gamma) \geq 0.$$

Proof. From the no-arbitrage condition

$$R(N^*(\gamma, \gamma'), \gamma) - R'(N'^*(\gamma', \gamma')) = 0.$$

So, given Lemmas C.1, C.2, and Corollary C.1, we can apply the implicit function theorem to obtain

$$\frac{dN^*}{d\gamma} = \frac{R_\gamma(N^*, \gamma) + R_N(N^*, \gamma) \frac{dN^*}{d\gamma}}{R'_{N'}(N'^*, \gamma')}. \quad (C.4)$$

Consider the case when $z^* > 0$. We have

$$R'(N'^*, \gamma') - \frac{\kappa}{2} (N^*(\gamma, \gamma') + N'^* - \omega)^2 - 1 = 0.$$

So, given Lemma C.1, and Corollary C.1, we obtain

$$\frac{dN'^*}{d\gamma} = \frac{\kappa M^* \frac{dN^*}{d\gamma}}{R'_{N'}(N'^*, \gamma') - \kappa M^*}. \quad (C.5)$$

Combining Eqs. (C.4) and (C.5) yields

$$\frac{dN^*}{d\gamma} = R_\gamma(N^*, \gamma) \times \underbrace{\left(\frac{\kappa M^* - R'_{N'}(N'^*, \gamma')}{R_N(N^*, \gamma) R'_{N'}(N'^*, \gamma') - \kappa M^* (R'_{N'}(N'^*, \gamma') + R_N(N^*, \gamma))} \right)}_{\equiv \xi^*(\gamma, \gamma')}.$$

which is well defined and continuous for all $\gamma, \gamma' \in (0, 1)$ and associated $\{N^*(\gamma, \gamma'), N'^*(\gamma', \gamma)\}$. Since $R_N(N^*, \gamma), R'_{N'}(N'^*, \gamma') < 0$ (Corollary C.1), and $\kappa M^* \geq 0$, this means $\xi^*(\gamma, \gamma') > 0$.

In the case where $z^* = 0$, we have $R(N^*, \gamma) - 1 = 0$, hence

$$\frac{dN^*}{d\gamma} = - \frac{R_\gamma(N^*, \gamma)}{R_N(N^*, \gamma)}.$$

The left hand side is equivalent to $R_\gamma(N^*, \gamma) \xi^*(\gamma, \gamma')$ when $M^* = 0$ (which is an implication of $z^* = 0$). Since $R_N(N^*, \gamma) < 0$ (Corollary C.1), $\xi^*(\gamma, \gamma') > 0$ when $M^* = 0$. Therefore the result holds for all $z^* \geq 0$.

To conclude, since $\xi^*(\gamma, \gamma') > 0$ is always positive, the sign of $\frac{dN^*}{d\gamma}$ is the same as the sign of $R_\gamma(N^*, \gamma)$. $\square$

Proposition 2. For all $\gamma, \gamma' \in (0, 1)$,

$$\frac{dN'^*}{d\gamma} = - \underbrace{\left(\frac{\kappa M^*}{\kappa M^* - R'_{N'}(N'^*, \gamma')} \right)}_{\equiv SP^*(\gamma, \gamma') \in [0, 1]} \frac{dN^*}{d\gamma}. \quad (C.6)$$

Proof. The result directly follows from Eq. (C.5) in the proof above, together with $R'_{N'}(N'^*, \gamma') < 0$ (Corollary C.1). If $z^* = 0$, then $M^* = 0$, and $\frac{dN'^*}{d\gamma} = 0$. $\square$

Theorem 1. Assume Regularity Condition 1 holds. Then, $\forall \gamma' \in (0, 1)$, there exists a $\hat{\gamma}(\gamma') > 0$ such that, $\forall \gamma \in (0, 1)$

$$\begin{cases} \frac{dN^*}{d\gamma} > 0 & ; \gamma < \hat{\gamma}(\gamma') \\ \frac{dN^*}{d\gamma} = 0 & ; \gamma = \hat{\gamma}(\gamma') \\ \frac{dN^*}{d\gamma} < 0 & ; \gamma > \hat{\gamma}(\gamma') \end{cases}$$

Proof. From Lemma 2 we know that the sign of $\frac{dN^*}{d\gamma}$ is the same as that of $R_\gamma(N^*(\gamma, \gamma'), \gamma)$. Based on this, the proof proceeds in steps to show the following:

- (i) $\frac{dR^*(\gamma, \gamma')}{d\gamma}$ exists and is continuous in γ , for all $\gamma, \gamma' \in (0, 1)$.
 - (ii) The sign of $R_\gamma(N^*(\gamma, \gamma'), \gamma)$ is the same as that of $\frac{dR^*(\gamma, \gamma')}{d\gamma}$ (and they are nil at the same γ).
 - (iii) At low values of γ , $R_\gamma(N^*(\gamma, \gamma'), \gamma)$ is positive, so $\frac{dN^*}{d\gamma} > 0$.
- Then, given regularity Condition 1, for each γ' , either $R_\gamma(N^*, \gamma)$ is positive for all $\gamma \in (0, 1)$, in which case $\hat{\gamma}(\gamma') = 1$ ensures that the Theorem holds, or there exists a $\hat{\gamma}(\gamma') < 1$, for which $R_\gamma(N^*(\hat{\gamma}(\gamma'), \gamma'), \hat{\gamma}(\gamma')) = 0$. But then, since regularity Condition 1 implies single crossing, it must be that $R_\gamma(N^*(\gamma, \gamma'), \gamma) < 0$, for all $\gamma > \hat{\gamma}(\gamma')$; ensuring that $\hat{\gamma}(\gamma')$ is indeed unique. And, by Lemma 2, it must be the case that $\frac{dN^*}{d\gamma}$ flips sign at $\gamma = \hat{\gamma}(\gamma')$, as per the Theorem.

We now proceed with the details of the three steps.

Step (i) Note that

$$\frac{dR^*(\gamma, \gamma')}{d\gamma} = R_\gamma(N^*, \gamma) + \frac{dN^*}{d\gamma} R_N(N^*, \gamma). \quad (C.7)$$

From Lemmas C.1, C.2, and 2 we know that all three elements of the right hand side exist and are continuous in γ .

Step (ii) From Lemma 2, we have

$$\frac{dN^*}{d\gamma} = R_\gamma(N^*, \gamma) \xi^*(\gamma, \gamma').$$

Substituting in Eq. (C.7) gives:

$$\frac{dR^*(\gamma, \gamma')}{d\gamma} = R_\gamma(N^*, \gamma) (1 + \xi^*(\gamma, \gamma') R_N(N^*, \gamma)).$$

But, we have

$$1 + \xi^*(\gamma, \gamma') R_N(N^*, \gamma) = \left(1 + \frac{\kappa M R_N - R'_{N'} R_N}{R_N R'_{N'} - \kappa M (R'_{N'} + R_N)} \right) \Big|_{N^*(\gamma, \gamma'), N^*(\gamma', \gamma)}$$

Hence

$$1 + \xi^*(\gamma, \gamma') R_N(N^*, \gamma) = \left(\frac{-\kappa M R'_{N'}}{-\kappa M R'_{N'} + R_N R'_{N'} - \kappa M R_N} \right) \Big|_{N^*(\gamma, \gamma'), N^*(\gamma', \gamma)},$$

and, given that $\kappa M^* \geq 0$, and that $R_N(N^*, \gamma) < 0$, $R'_{N'}(N^*, \gamma') < 0$ (Corollary C.1), it is the case that

$$1 + \xi^*(\gamma, \gamma') R_N(N^*, \gamma) \in [0, 1],$$

and, therefore, that

$$\text{sign} \left(\frac{dR^*(\gamma, \gamma')}{d\gamma} \right) = \text{sign} (R_\gamma(N^*, \gamma)).$$

Step (iii) We start from

$$R_\gamma(N, \gamma) = \frac{-1}{\gamma^2} \int_{A^0(N, \gamma)}^{A^H} \left(\alpha^2 A \left(\frac{N}{\gamma} \right)^{\alpha-1} - 1 \right) g(A) dA. \quad (C.8)$$

and consider the limit as γ tends to 0 from above. It is positive if

$$\lim_{\gamma \rightarrow 0^+} \int_{A^0}^{A^H} \left(\alpha^2 A \left(\frac{N}{\gamma} \right)^{\alpha-1} - 1 \right) g(A) dA < 0.$$

It turns out that it must be the case in equilibrium. To see why, consider X such that $\alpha(X)^{\alpha-1} = 1$. At such X , as γ tends to 0, banks make strictly positive profits (lending just breaks even in expectation, but banks have no equity, so they only pick the upside, i.e., $A^0 > 0$). This cannot be true in equilibrium. Instead, it must be the case that $\alpha(X^*)^{\alpha-1} < 1$. Since $\alpha < 1$, this is also true for $\alpha^2(X^*)^{\alpha-1} < 1$. So, $R_\gamma(N^*, \gamma)$ is positive at low enough values of γ . $\square$

The next two lemmas will be useful for subsequent proofs.

Lemma C.3. (a) $\frac{dX^*}{d\gamma} < 0$ and (b) $\frac{dN^*}{d\gamma} < X^*$.

Proof. First, we show that $\frac{dX^*}{d\gamma} < 0$. Using a change of variable, we have

$$\check{R}(X^*, \gamma) - (1 + \check{z}(X^*, \gamma, \gamma')) = 0.$$

From which we get

$$\frac{dX^*}{d\gamma} = - \frac{\check{R}_\gamma - \check{z}_\gamma}{\check{R}_X - \check{z}_X}. \quad (C.9)$$

Because of the diminishing marginal product of physical capital, we have $\check{R}_X < 0$. Furthermore, even though R_γ can be positive (as we show in Section 3), $\check{R}_\gamma$ cannot: an increase in γ keeping X constant can only decrease the return on bank capital, as this decreases the value of the implicit subsidy from the government guarantee without altering the gross revenues through changes in aggregate lending.

If $z^* = 0$, we have $\check{z}_\gamma, \check{z}_X = 0$. So both the numerator and the denominator in Eq. (C.9) are strictly negative and, therefore, $\frac{dX^*}{d\gamma} < 0$.

If $z^* > 0$, we have that $\check{z}(X^*, \gamma, \gamma') = \frac{\kappa}{2}(X^*\gamma + N^*(\gamma', \gamma) - \omega)^2$, so $\check{z}_X > 0$: keeping γ constant, an increase in X^* other variable increases N^* and therefore the equilibrium cost of bank capital. So the denominator in Eq. (C.9) is still negative.

Now

$$\check{z}_\gamma = \kappa(X^*\gamma + N^*(\gamma', \gamma) - \omega) \left(X^* + \frac{dN^*}{d\gamma} \right).$$

Appealing to Proposition 2, we know that

$$\frac{dN^*}{d\gamma} = -SP^* \frac{dN^*}{d\gamma} = -SP^* \left(X^* + \frac{dX^*}{d\gamma} \right).$$

Hence, we can write

$$\check{z}_\gamma = \kappa(X^*\gamma + N^*(\gamma', \gamma) - \omega) \left((1 - SP^*)X^* - \gamma SP^* \frac{dX^*}{d\gamma} \right).$$

Substituting this expression into Eq. (C.9) and rearranging, we have, when $z^* > 0$, the following expression:

$$\frac{dX^*}{d\gamma} = - \left(\frac{\check{R}_\gamma - \kappa(X^*\gamma + N^*(\gamma', \gamma) - \omega)(1 - SP^*)X^*}{\check{R}_X - \check{z}_X} \right) \times \left(1 - \frac{\gamma SP^*}{\check{R}_X - \check{z}_X} \right)^{-1}. \quad (C.10)$$

It is the case that $\kappa(X^*\gamma + N^*(\gamma', \gamma) - \omega)(1 - SP^*)X^*$ is strictly positive when $z^* > 0$. Since $(\check{R}_X - \check{z}_X)$ and $\check{R}_\gamma$ are both negative, the first parenthesis is strictly positive. And $(\check{R}_X - \check{z}_X) < 0$ also implies that the second parenthesis is positive. Hence, $\frac{dX^*}{d\gamma} < 0$; which proves (a).

Then, since $N^* = \gamma X^*$, we have $\frac{dN^*}{d\gamma} \equiv X^* + \gamma \frac{dX^*}{d\gamma}$, and $\frac{dX^*}{d\gamma} < 0$ implies that $\frac{dN^*}{d\gamma} < X^*$; which proves (b). $\square$

Lemma C.4. (a) Under Condition 2, γ^{nash} and γ^{col} exist and are unique.

(b) Write the regulator's objective as $\pi^*(\gamma, \gamma') = V(N^*(\gamma, \gamma'), \gamma)$. Use a change of variable to define $\tilde{V}(X(\gamma, \gamma'), N^*(\gamma, \gamma')) \equiv V(N^*(\gamma, \gamma'), \gamma)$. Then, $\tilde{V}_N(X, N) > 0$ ensures that $V_N(\gamma^{\text{nash}}, \gamma^{\text{nash}}) > 0$.

Proof. (a) is straightforward. (b) First, we prove $V_N^{\text{nash}} > 0$. Given γ' , the competitive regulator's first order condition is:

$$\pi^*_\gamma(\gamma, \gamma') = V_\gamma(N^*, \gamma) + \frac{dN^*}{d\gamma} V_N(N^*, \gamma) = 0. \quad (C.11)$$

With the change of variable, we obtain:

$$\tilde{V}_N(X^*, N^*) \equiv \frac{V_\gamma(N^*, \gamma)}{X^*} + V_N(N^*, \gamma).$$

Substituting $V_N(N^*, \gamma)$ in the first order condition gives:

$$V_\gamma(N^*, \gamma) + \frac{dN^*}{d\gamma} \left(\tilde{V}_N(X^*, N^*) - \frac{V_\gamma(N^*, \gamma)}{X^*} \right) = 0$$

or

$$\frac{dN^*}{d\gamma} \tilde{V}_N(X^*, N^*) = V_\gamma(N^*, \gamma) \left(\frac{1}{X^*} \frac{dN^*}{d\gamma} - 1 \right).$$

From Lemma C.3, we know that $\left(\frac{1}{X^*} \frac{dN^*}{d\gamma} - 1\right) < 0$. So, we have that $\tilde{V}_N(X^*, N^*) > 0$ implies that $V_\gamma(N^*, \gamma)$ and $\frac{dN^*}{d\gamma}$ have opposite signs along the best response function. But then, from the first order condition (C.11), it must be that $V_N(N^*, \gamma) > 0$. If this is true along the best response function, it is true at the Nash equilibrium: $V_N^{\text{nash}} > 0$.

For completeness, we can also prove that $V_N^{\text{col}} > 0$. Given γ' , the collaborative regulator first order condition for γ is:

$$V_\gamma(N^*, \gamma) + (1 - \text{SP}^*) \frac{dN^*}{d\gamma} V_N(N^*, \gamma) = 0. \quad (\text{C.12})$$

Following the same steps as above, we get

$$\frac{dN^*}{d\gamma} \tilde{V}_N(X^*, N^*) = V_\gamma(N^*, \gamma) \left((1 - \text{SP}^*) \frac{1}{X^*} \frac{dN^*}{d\gamma} - 1 \right).$$

And noting that $\text{SP}^* \in [0, 1]$ yields, by the same logic as with the V_N^{nash} , the result: $V_N^{\text{col}} > 0$. $\square$

Proposition 3. Assume Conditions 1 and 2 hold. Then, there are three cases

$$\begin{cases} \gamma^{\text{col}} < \gamma^{\text{nash}} < \bar{\gamma} & \text{and } \frac{dN^{\text{col}}}{d\gamma} > 0 \\ \gamma^{\text{col}} = \gamma^{\text{nash}} = \bar{\gamma} & \text{and } \frac{dN^{\text{col}}}{d\gamma} = 0 \\ \gamma^{\text{col}} > \gamma^{\text{nash}} > \bar{\gamma} & \text{and } \frac{dN^{\text{col}}}{d\gamma} < 0, \end{cases}$$

where if $\bar{\gamma} \in (0, 1)$, it is the fixed point $\bar{\gamma} = \hat{\gamma}(\bar{\gamma})$, with $\hat{\gamma}(\cdot)$ defined in Theorem 1, otherwise $\bar{\gamma} = 1$.

Proof. The proof proceeds in three steps to show the following

(i) We prove that if a fixed point $\bar{\gamma} = \hat{\gamma}(\bar{\gamma})$ exists for values $\gamma \in (0, 1)$, it is unique. So, the sign of any $\frac{dN^*}{d\gamma}$ at any symmetric choice $\gamma = \gamma'$ only depends on whether $\gamma \leq \bar{\gamma}$. Hence, the sign of $\frac{dN^{\text{col}}}{d\gamma}$ depends on $\gamma \leq \bar{\gamma}$, as per Theorem 1.

(ii) We derive first order conditions (Nash and collaborative). And we construct a function $L(\gamma)$ that gives the collaborative marginal benefit of a change in γ , at a point where $\gamma' = \gamma$.

(iii) We evaluate $L(\gamma)$ at γ^{nash} and $\bar{\gamma}$, and show by contradiction that only the three cases can occur.

Finally, for completeness, we also provide the derivations for Eq. (13) in the text.

Step (i)

We prove that if a fixed point $\bar{\gamma} = \hat{\gamma}(\bar{\gamma})$ exists for values $\gamma \in (0, 1)$, it is unique. Denote Γ a common capital requirement for Home and Foreign.

Our strategy is to show that the properties of $R(N^*(\gamma, \gamma'), \gamma')$ with respect to γ used in the proof of Theorem 1 also apply to $R(N^*(\Gamma, \Gamma), \Gamma)$ with respect to Γ . By the same reasoning as in Theorem 1 this is sufficient to prove that either there exists a unique $\bar{\gamma} \in (0, 1)$ that maximises $R(N^*(\Gamma, \Gamma), \Gamma)$, or that $\frac{R(N^*(\Gamma, \Gamma), \Gamma)}{d\Gamma} > 0$ at $\Gamma = 1$ and $R(N^*(\Gamma, \Gamma), \Gamma) < R(N^*(1, 1), 1)$ for all $\Gamma < 1$.

At $\{\Gamma, \Gamma\}$ we have the equilibrium condition

$$R(N^*, \Gamma) - \frac{\kappa}{2} (2N^* - \omega)^2 - 1 = 0.$$

$$\text{We then obtain } \frac{dN^*}{d\Gamma} = \frac{R_\Gamma(N^*, \Gamma)}{2\kappa M^* - R_N(N^*, \Gamma)}.$$

So

$$\frac{dR(N^*(\Gamma, \Gamma), \Gamma)}{d\Gamma} = R_\gamma^* + \frac{dN^*}{d\Gamma} R_N^* = \frac{2\kappa M^*}{2\kappa M^* - R_N^*} R_\gamma^*.$$

From here it is obvious that the first two properties in the proof of Theorem 1 apply to $R(N^*(\Gamma, \Gamma), \Gamma)$. The third property, that $R_\gamma^*(N^*(\Gamma, \Gamma), \Gamma)$ is positive for low values of Γ , applies following the reasoning in the proof in Theorem 1. Hence, the sign of $\frac{dN^*}{d\gamma}$ at any symmetric choice $\gamma = \gamma'$ only depends on whether $\gamma \leq \bar{\gamma}$.

Step (ii)

We endowed the regulator with the objective

$$\pi^*(\gamma, \gamma') \equiv \tilde{V}(X^*(\gamma, \gamma'), N^*(\gamma, \gamma'))$$

$$= (X^*(\gamma, \gamma'))^\alpha - X^*(\gamma, \gamma') - \lambda S(X^*(\gamma, \gamma'), N^*(\gamma, \gamma')).$$

This can be reexpressed as:

$$\pi^*(\gamma, \gamma') \equiv V(N^*(\gamma, \gamma'), \gamma) \equiv \left(\frac{N^*}{\gamma}\right)^\alpha - \frac{N^*}{\gamma} - \lambda S\left(\frac{N^*}{\gamma}, N^*\right).$$

The Nash first order condition is

$$V_\gamma^{\text{nash}} + V_N^{\text{nash}} \frac{dN^{\text{nash}}}{d\gamma} = 0, \quad (\text{C.13})$$

and the collaborative first order condition (13) is

$$V_\gamma^{\text{col}} + V_N^{\text{col}} \frac{dN^{\text{col}}}{d\gamma} + V_{N'}^{\text{col}} \frac{dN'^{\text{col}}}{d\gamma} = 0.$$

Define $L(\gamma)$ as the function equal to the left hand side of the collaborative first order condition (13) when both countries play γ . We can rewrite the collaborative first order condition (13)

$$L(\gamma^{\text{col}}) \equiv \pi_\gamma^*(\gamma^{\text{col}}, \gamma^{\text{col}}) + \pi_{\gamma'}^*(\gamma^{\text{col}}, \gamma^{\text{col}}) = 0.$$

Substituting the function $L(\gamma)$ into condition (C.13), we obtain

$$L(\gamma^{\text{nash}}) - \underbrace{V_{N'}^{\text{nash}} \frac{dN'^{\text{nash}}}{d\gamma}}_{\equiv \pi_{\gamma'}^*(\gamma^{\text{nash}}, \gamma^{\text{nash}})} \equiv \pi_\gamma^*(\gamma^{\text{nash}}, \gamma^{\text{nash}}) = 0. \quad (\text{C.14})$$

Now, from the properties of the shortfall function we have $\tilde{V}_N > 0$. Hence, from Lemma C.4 $V_{N'}^{\text{nash}} > 0$.

Step (iii)

We now confirm that only three different cases are possible.

First, assume $\gamma^{\text{nash}} < \gamma^{\text{col}}$. Condition 2 guarantees $L_\gamma(\gamma) < 0$. So $L(\gamma^{\text{nash}}) > L(\gamma^{\text{col}}) = 0$ as increasing γ from γ^{nash} towards γ^{col} would increase social welfare. Since $V_{N'}^{\text{nash}} > 0$, for (C.14) to hold, it must be that $\frac{dN^{\text{nash}}}{d\gamma} < 0$. Hence, $\gamma^{\text{nash}} < \gamma^{\text{col}}$ is only compatible with $\gamma^{\text{col}} > \bar{\gamma}$. In addition, $\gamma^{\text{col}} > \bar{\gamma}$ implies $L(\gamma) > 0$ for all $\gamma \leq \bar{\gamma}$. But $\gamma \leq \bar{\gamma}$ also implies $\frac{dN^*}{d\gamma} \geq 0$ (by step (i) above), so it must be that $\pi_\gamma^*(\gamma, \gamma) > 0$, for all $\gamma \leq \bar{\gamma}$, and so there is no solution to Eq. (C.14), and hence no Nash equilibrium, in the range $\gamma \leq \bar{\gamma}$. This rules out $\gamma^{\text{nash}} < \bar{\gamma}$. Hence, $\gamma^{\text{nash}} < \gamma^{\text{col}} \Leftrightarrow \bar{\gamma} < \gamma^{\text{nash}}$.

Second, assume $\gamma^{\text{col}} > \gamma^{\text{nash}}$. Then, $L(\gamma) < L(\gamma^{\text{col}}) = 0$ as decreasing γ from γ^{nash} towards γ^{col} would increase social welfare. Since $V_{N'}^{\text{nash}} > 0$, for (C.14) to hold, it must be that $\frac{dN^{\text{nash}}}{d\gamma} > 0$. Hence, $\gamma^{\text{nash}} > \gamma^{\text{col}}$ is only compatible with $\gamma^{\text{col}} < \bar{\gamma}$. In addition, $\gamma^{\text{col}} < \bar{\gamma}$ implies $L(\gamma) < 0$ for all $\gamma \geq \bar{\gamma}$. But $\gamma \geq \bar{\gamma}$ also implies $\frac{dN^*}{d\gamma} \leq 0$ (by step (i) above), so it must be that $\pi_\gamma^*(\gamma, \gamma) < 0$ for all $\gamma \geq \bar{\gamma}$, and so there is no solution to Eq. (C.14), and hence no Nash equilibrium, in the range $\bar{\gamma} \leq \gamma$. This rules out $\gamma^{\text{nash}} > \bar{\gamma}$. Hence, $\gamma^{\text{nash}} > \gamma^{\text{col}} \Leftrightarrow \bar{\gamma} > \gamma^{\text{nash}}$.

Finally, assume $\gamma^{\text{col}} = \bar{\gamma}$. Then the Nash (C.13) and Collaborative (13) first order conditions coincide, and it must be that $\gamma^{\text{col}} = \gamma^{\text{nash}}$. Which concludes the proof.

Additional derivation

We want to show that the collaborative first order condition can be written

$$\pi_\gamma^*(\gamma, \gamma') + \underbrace{\left(\frac{\text{NPV}^*}{\gamma'} - \lambda S_{N'}^* \right) \frac{dN'^*}{d\gamma}}_{\equiv \pi_{\gamma'}^*(\gamma', \gamma')} = 0,$$

as it is in the main text (13).

By construction, the Collaborative FOC is

$$\pi_\gamma^*(\gamma, \gamma') + \pi_{\gamma'}^*(\gamma', \gamma) = 0.$$

Let us derive the second term

$$\pi_{\gamma'}^*(\gamma', \gamma) = \left(\alpha (X'^*)^{\alpha-1} - 1 - \lambda S_{X'}^* \right) \left(\frac{dX'^*}{d\gamma} \right) - \lambda S_{N'}^* \left(\frac{dN'^*}{d\gamma} \right).$$

Since $X'^* = N'^* / \gamma'$, we have

$$\frac{dX'^*}{d\gamma} = \frac{1}{\gamma'} \frac{dN'^*}{d\gamma}.$$

Recalling $NPV^* = \alpha (X^*)^{\alpha-1} - 1 - \lambda S_X^*$, we get

$$\pi_{\gamma'}^*(\gamma') = (NPV^*) \left(\frac{1}{\gamma'} \frac{dN^*}{d\gamma'} \right) - \lambda S_{N'}^* \left(\frac{dN^*}{d\gamma'} \right)$$

$$\pi_{\gamma'}^*(\gamma') = \left(\frac{NPV^*}{\gamma'} - \lambda S_{N'}^* \right) \frac{dN^*}{d\gamma'}$$

which corresponds to the above. $\square$

Proposition 4. Assume $\kappa \rightarrow \infty$, [Conditions 1](#) and [2](#) hold, and $g(A)$ is continuous for all $A^0 > 0$ with full support over $[0, A^H]$. Then $\forall \alpha, \lambda, \exists \bar{\omega} \leq \bar{\omega} < 2\alpha^{\frac{\alpha}{1-\alpha}}$:

$$\begin{cases} \gamma^{\text{col}} < \gamma^{\text{nash}} < \bar{\gamma} & ; \omega > \bar{\omega} \\ \gamma^{\text{col}} = \gamma^{\text{nash}} = \bar{\gamma} & ; \omega = \bar{\omega} \\ \gamma^{\text{col}} > \gamma^{\text{nash}} > \bar{\gamma} & ; \omega < \bar{\omega}. \end{cases}$$

Proof. To begin let us first outline our strategy for the proof.

1. Step 1

- Show $N^{\text{col}} = \omega/2$
- Show X^{col} is increasing in ω .
- Show, given $\omega < 2\alpha^{\frac{\alpha}{1-\alpha}}$, $\gamma^{\text{col}} = \frac{\omega/2}{X^{\text{col}}} < 1$

2. Step 2

- Show $N^*(\bar{\gamma}, \bar{\gamma}) \equiv \bar{N} = \omega/2$.
- Show for all $\omega/2 \geq \alpha^{\frac{\alpha}{1-\alpha}}$, $\bar{X} = \alpha^{\frac{\alpha}{1-\alpha}}$. Given $\gamma^{\text{col}} < 1$, we have $X^{\text{col}} > \omega/2 \geq \alpha^{\frac{\alpha}{1-\alpha}} = \bar{X}$
- Show for all $\omega < \alpha^{\frac{\alpha}{1-\alpha}}$, $\bar{X}$ is decreasing in ω . Since X^{col} is increasing in ω , there are two cases.

- If $\lim_{\omega \rightarrow 0} X^{\text{col}} > \lim_{\omega \rightarrow 0} \bar{X}$ then $\bar{\omega} = 0$.
- Otherwise, there exists an $0 < \bar{\omega} < \alpha^{\frac{\alpha}{1-\alpha}}$ such that $\bar{X} = X^{\text{col}}$ and $\bar{X} > X^{\text{col}}$ for all $\omega < \bar{\omega}$.

3. Step 3: Given $N^{\text{col}} = \bar{N} = \omega/2$, we have $X^{\text{col}} \leq \bar{X} \Leftrightarrow \gamma^{\text{col}} \geq \bar{\gamma}$

Step 1

We start from [\(14\)](#)

$$\alpha (X^{\text{col}})^{\alpha-1} = 1 + \lambda \underbrace{\int_0^{A^0} (1 - \alpha^2 A (X^{\text{col}})^{\alpha-1}) g(A) dA}_{\equiv S_X^*} \quad (\text{C.15})$$

where $A^0 = \frac{X^{\text{col}} - N^{\text{col}}}{\alpha (X^{\text{col}})^{\alpha}}$ for $X^{\text{col}} > N^{\text{col}}$ and $A^0 = 0$ otherwise.

(a) First, note that $S_X^* \geq 0$. If $S_X^* = 0$, then $X^{\text{col}} = \alpha^{\frac{\alpha}{1-\alpha}}$ and this is an upper bound (it cannot be optimal for regulators to allow for loans that have a negative NPV even before taking deadweight losses into account). Now, assume $N^{\text{col}} < \omega/2$. Given we have imposed: $\omega < 2\alpha^{\frac{\alpha}{1-\alpha}}$, this means that $X^{\text{col}} < \alpha^{\frac{\alpha}{1-\alpha}}$. But then, loans must have strictly positive NPV, which means that the return on bank equity must be strictly greater than 1. But, given [Proposition 1](#), this implies $z^* > 0$, which is a contradiction with $N^{\text{col}} < \omega/2$. Hence $N^{\text{col}} = \omega/2$.

(b) The left side of [\(C.15\)](#) is infinite at $X^{\text{col}} = 0$, decreasing in X^{col} and tends to $\alpha < 1$ as X^{col} grows large. Now $S_X^* = 0$ for $X^{\text{col}} \leq \omega/2$ and is strictly increasing for $X^{\text{col}} > \omega/2$ (the integrand is always positive for $A \leq A^0$). Hence, there exists a unique X^{col} that solves [\(14\)](#). Denote such implicit function $X^{\text{col}}(\omega)$.

Then,

$$S_{X\omega}^* = \underbrace{\frac{\partial A^0}{\partial \omega}}_{\leq 0} \underbrace{\left(1 - \alpha^2 A^0 (X^{\text{col}})^{\alpha-1} \right)}_{> 0} f(A^0).$$

So, $\frac{dX^{\text{col}}}{d\omega} \geq 0$, strictly if $A^0 > 0$ (Since $g(A)$ is continuous for all $A^0 > 0$, and has full support over $[0, A^H]$). Next, we show that $\gamma^{\text{col}} < 1$, so that $\frac{dX^{\text{col}}}{d\omega} > 0$ for all $\omega < \alpha^{\frac{\alpha}{1-\alpha}}$.

(c) Finally, we have $\gamma^{\text{col}} = \frac{\omega/2}{X^{\text{col}}}$. Assume $X^{\text{col}} \leq \omega/2$, then $A^0 = 0$, but then $X^{\text{col}} = \alpha^{\frac{\alpha}{1-\alpha}} > \omega/2$; a contradiction. Hence, $\gamma^{\text{col}} < 1$.

Step 2

(a) Define $\bar{N} \equiv N^*(\bar{\gamma}, \bar{\gamma})$. If $\bar{N} < \omega/2$, $R(\bar{N}, \bar{\gamma}) = 1$. But from Step 1(a) $R(\omega/2, \gamma^{\text{col}}) > 1$. Hence, $\bar{N} < \omega/2$ cannot be the case, as $\bar{\gamma}$ would not maximise banks RoE. Hence, we must have $R(\omega/2, \bar{\gamma}) > 1$ and $\bar{N} = \omega/2$.

(b) Define $\bar{X}(\omega)$ the value for X that solves $R_{\gamma}(\omega/2, \frac{\omega/2}{\bar{X}}) = 0$. That is,

$$\int_{A^0}^{A^H} (\alpha^2 A \bar{X}^{\alpha-1} - 1) g(A) dA = 0 \quad (\text{C.16})$$

where $A^0 = \frac{\bar{X} - \omega/2}{\alpha (\bar{X})^{\alpha}}$ for all $\bar{X} > \omega/2$ and $A^0 = 0$ otherwise.

Condition [\(C.16\)](#) can be rewritten:

$$\alpha^2 \bar{X}^{\alpha-1} + \underbrace{\int_0^{A^0} (1 - \alpha^2 A \bar{X}^{\alpha-1}) g(A) dA}_{\equiv S_X^*} = 1$$

Here, S_X^* reflects the implicit subsidy from government guarantees. As noted above $S_X^* \geq 0$. Hence, the maximum value $\bar{X}$ can possibly take corresponds to $S_X^* = 0$. This happens when $\bar{X}(\omega) < \omega/2$. Then, $A^0 = 0$ and $\alpha (\bar{X}(\omega))^{\alpha-1} = \frac{1}{\alpha}$, which yields, $\bar{X}(\omega) = \alpha^{\frac{\alpha}{1-\alpha}}$ (and corresponds to X^{mon} in [Section 3.2](#)).

So, for all $\omega/2 \geq \alpha^{\frac{\alpha}{1-\alpha}}$, $\bar{X}(\omega) = \alpha^{\frac{\alpha}{1-\alpha}}$ and, given $\gamma^{\text{col}} < 1$, we have $X^{\text{col}}(\omega) > \omega/2 \geq \alpha^{\frac{\alpha}{1-\alpha}} = \bar{X}(\omega)$. Equivalently, $\bar{\gamma} = (\omega/2) / (\alpha^{\frac{\alpha}{1-\alpha}}) > 1 > \gamma^{\text{col}}$.

(c) Now, consider $\omega < \alpha^{\frac{\alpha}{1-\alpha}}$. For all such ω , we know that decreasing ω increases $\bar{X}$ and decreases X^{col} . So there are two cases.

If

$$\lim_{\omega \rightarrow 0} \alpha (\bar{X}(\omega))^{\alpha-1} > 1 + \lim_{\omega \rightarrow 0} \lambda \int_0^{A^0} (1 - \alpha^2 A (\bar{X}(\omega))^{\alpha-1}) g(A) dA,$$

this means that $\lim_{\omega \rightarrow 0} X^{\text{col}}(\omega) > \lim_{\omega \rightarrow 0} \bar{X}(\omega)$. Then only $\gamma^{\text{col}} < \gamma^{\text{nash}} < \bar{\gamma}$ can arise. Since $\omega > 0$, setting $\bar{\omega} = 0$ ensures that the proposition holds.

Otherwise, there exists an $0 < \bar{\omega} < \alpha^{\frac{\alpha}{1-\alpha}}$ such that $\bar{X}(\bar{\omega}) = X^{\text{col}}(\bar{\omega})$ and $\bar{X}(\omega) > X^{\text{col}}(\omega)$ for all $\omega < \bar{\omega}$.

Step 3

Given $N^{\text{col}} = \bar{N} = \omega/2$, we have $X^{\text{col}} \leq \bar{X} \Leftrightarrow \gamma^{\text{col}} \geq \bar{\gamma}$, which concludes the proof. $\square$

Proposition 6. For any objective $\pi^*(\gamma, \gamma') = V(N^*(\gamma, \gamma'), \gamma)$, if $V_N(N^{\text{nash}}, \gamma^{\text{nash}}) > 0$, then [Proposition 3](#) holds.

Proof. This follows directly from the proof of [Proposition 3](#). $\square$

Proposition 7. Define $\tilde{V}(X, N) \equiv V(N, \gamma = N/X)$. Then $\tilde{V}_N(X, N) > 0, \forall N < X$ is a sufficient condition for [Proposition 3](#) to hold under general objective [\(B.2\)](#).

Proof. From [Proposition 6](#), we just need to prove that $V_N(N^{\text{nash}}, \gamma^{\text{nash}}) > 0$ if $\tilde{V}_N(X, N) > 0$. This is true from [Lemma C.4](#). $\square$

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